Integrated Framework for Long-term Surgery Planning Management under Uncertainty: From Strategic to Tactical Planning

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Abstract

Abstract text.

Keywords:

1. Related Works

Surgery scheduling problems have traditionally been classified into three planning levels with different sources of uncertainty: strategic, tactical, and operational (Rahimi and Gandomi, 2020; Cardoen et al., 2010; Wang et al., 2021; Melo and Fogliatto, 2025). At the strategic level, long-term decisions define the desired split of the capacity into specific surgical specialities and specify patient volumes of different medical specialties to be admitted over the planning horizon (Burdett et al., 2023) — a process commonly referred to as Case Mix Planning (CMP). Different objectives have been introduced at this level, such as maximizing profit (Fügener, 2015; Ma and Demeulemeester, 2013) or minimizing long-term costs (Hof et al., 2015; McRae and Brunner, 2020). The tactical level defines a cyclic schedule (often weekly) commonly known as Master Surgery Schedule (MSS) (Bovim et al., 2020; Siqueira et al., 2018; Carneiro et al., 2025). The MSS step allocates time blocks in operating theatres to surgical teams and is reviewed in a medium-term horizon (e.g., monthly or quarterly). Finally, the operational level focuses on the daily operation and derives a Surgical Case Schedule (SCS), which generally assigns individual patients to time blocks previously allocated to specific surgical teams at the tactical level (Vieira et al., 2025; Harris and Claudio, 2022).

Despite the large body of literature in surgery scheduling, there are significant gaps to be bridged. For example, inter-level integration remains limited in scope, often focusing on short-term alignment between surgeries and post-surgical bed occupation, and generally relying on deterministic approaches (Melo and Fogliatto, 2025; Wang et al., 2021; Siqueira et al., 2018). Furthermore, regardless of the planning level, strategies with a focus on guaranteeing waiting time requirements for patients still remain to be proposed. This is probably due to the complex propagation of uncertainty over different planning levels, which demands hybrid approaches that combine mathematical programming, stochastic modelling and queuing theory.

The inherent interconnection between the strategic and tactical levels has motivated unified frameworks that link CMP and MSS decisions (Melo and Fogliatto, 2025). These often adopt sequential or joint optimisation approaches where case mix decisions act as the input to surgery scheduling models (Ma and Demeulemeester, 2013; Fügener, 2015). Modelling approaches typically consist of Mixed Integer Programming (MIP) formulations that focus on maximising hospital profit subject to Operating Theatre (OT) and bed capacity constraints. While these frameworks acknowledge uncertainty, it is generally confined to downstream elements such as length of stay or ward occupancy. Although modelling post-surgical congestion is relevant, such partial treatments remain short-sighted when not complemented by uncertainty in surgery durations and, crucially, by an explicit modelling of dynamic waiting lists across surgical specialities, which constitute the starting point of the entire planning process.

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The perspective can be broadened by incorporating additional uncertainty sources and dynamic elements. For instance, Liu et al. (2019) model elective admissions through a Markov Decision Process (MDP), highlighting the role of waiting-list size in guiding strategic decisions. Despite its conceptual value, the framework disregards essential practical constraints, such as surgeon availability, block scheduling and case-mix heterogeneity, thereby limiting practical applicability. Similarly, McRae and Brunner (2020) demonstrate that explicitly modelling uncertainty and selecting appropriate aggregation levels can significantly improve CMP outcomes and enable joint CMP-MSS optimisation. Nevertheless, waiting-list dynamics remain outside of the modelling scope, implying no full integration of strategic and tactical decisions. Furthermore, their reliance on Sample Average Approximation (SAA) leads to representations that may understate distributional variability. These studies underscore the value of integrating CMP and MSS, yet they also reveal persistent shortcomings: uncertainty is represented only selectively across planning levels, waiting-list requirements are rarely modelled explicitly, and key operational constraints required for real-world implementation remain overlooked.

Uncertainty in surgery duration is another important issue that, when disregarded, may result in unpredictable idle or overtime in OTs (Shehadeh and Padman, 2022). Whilst overtime is costly and can lead to cancellations that compromise the quality of care, idle time reflects an inadequate use of OT capacity (Shehadeh and Padman, 2022; Harris and Claudio, 2022; Kayvanfar et al., 2025). With the aim of maximising hospital revenue while preventing excessive overtime, Barz and Rajaram (2015) propose a tactical patient admission and scheduling problem that derives a CMP under multiple resource constraints and stochastic clinical progression. The approach utilises a newsvendor heuristic to reserve capacity for emergency arrivals, and an MDP model to select the patient cohort to be admitted whilst considering probabilistic transitions between clinical states. The model, however, excludes the possibility of cancellations or postponements driven by overtime, as it assumes that the hospital always disposes of the resources to treat each admitted patient. Moreover, by treating resource consumption as deterministic, the framework effectively ignores uncertainty in surgery duration — one of the main drivers of operational instability. As a result, the model underestimates the practical interplay between stochastic surgery durations, cancellations and overtime, limiting its applicability to integrated CMP-MSS decision-making.

Furthermore, Zhang et al. (2020) study a tactical surgery planning with the explicit aim of mitigating overtime risk when allocating elective patients to OTs over an upcoming planning horizon. Their model accounts for stochastic surgery durations and adopts a risk-averse objective that jointly captures both the probability and the expected magnitude of overtime, extending earlier approaches that consider only one of these dimensions (Shylo et al., 2013; Rath et al., 2017). Although they derive the durations from real-world data, eventually they approximate it through a set of discrete scenarios, limiting the full distributional representation. A similar treatment appears in Shehadeh et al. (2026), which introduces a Distributionally Robust Optimisation (DRO) approach to address uncertainty in surgery durations. While their approach minimises worst-case expected cost over an ambiguity set defined by known moments and support, it likewise relies on scenario-based representations. Accurately capturing surgery-duration variability requires a large number of scenarios, substantially increasing computational burden and potentially undermining the practical advantages of DRO and related methods.

These limitations regarding the integration of overtime management and its associated cancellation risks into surgical planning frameworks highlight important research gaps. There remains a need for models that explicitly incorporate cancellation dynamics induced by overtime and that provide a more thorough representation of surgery-duration uncertainty beyond scenario-based approximations. A key challenge which we tackle in this paper is to develop a modular approach to capture full distributional characteristics to build a planning model and use its output to feed an integrated CMP-MSS approach — i.e., not solving an entire model at once.

An additional and often underexplored dimension in integrated surgical planning concerns the incorporation of practical constraints inherent to real-world healthcare settings. Certain surgical activities, such as organ transplantation, impose rigid scheduling requirements that are difficult to reconcile with standard optimisation assumptions (da Silva et al., 2025). Furthermore, public hospitals in developing countries (e.g., Brazil) operate under severe resource constraints, high and volatile demand, and heterogeneous case mixes, all of which exacerbate the complexity of surgical planning (Carneiro et al., 2025; Siqueira et al., 2018; da Silva et al., 2025). These contextual factors challenge the scalability and transferability of many optimisation-based approaches, which often presume stable resources, predictable demand and homogeneous patient populations. Indeed, as noted by Melo and Fogliatto (2025), much of the existing literature is calibrated to relatively small hospitals in developed regions, limiting its relevance for large, resource-constrained systems. Ignoring such operational realities risks producing models that are theoretically stable but difficult to implement in practice, underscoring the need for scheduling frameworks that explicitly accommodate institutional, organisational and contextual constraints.

This paper addresses the identified gaps in the surgical planning literature by proposing an integrated framework that jointly links strategic and tactical decision levels while explicitly accounting for key sources of uncertainty. The framework captures dynamic surgical demand through explicit waiting-list modelling, enabling the enforcement of waiting-time requirements and long-term queue stability (Shortle et al., 2018), and incorporates stochastic surgery durations to properly balance cancellation, idle time, and overtime risks for each surgical session. The proposed approach is grounded in a real-world application at the Clinical Hospital of the University of Campinas (HC-Unicamp) in Brazil, a high-complexity public hospital that provides detailed operational data and a challenging planning environment (HC - Unicamp, 2024b).

The main contributions of this work are as follows:

- A stochastic waiting-list management strategy based on an (R, Q) control policy (Axsäter, 2015) that ensures compliance with waiting-time targets while maintaining long-term queue stability across surgical specialties;
- A cancellation risk management mechanism based on a newsvendor formulation (Arrow et al., 1951), which determines optimal time buffers for the last surgery in each session under duration uncertainty and defines probability distributions for the number of planned surgeries that can be accommodated without incurring overtime;
- A MIP model for operating theatre allocation that: (i) incorporates practical constraints, including those associated with transplant surgeries, while remaining computationally tractable; and (ii) explicitly incorporates the waiting-list management strategy and cancellation risk considerations from the previous items;
- An empirical validation of the integrated framework with data from a large, resource-constrained public hospital, demonstrating its operational relevance and implementability.

By jointly addressing relevant sources of uncertainty within a unified framework, this work introduces an innovative and integrated surgical planning approach that advances the literature by bridging multiple relevant gaps. The approach also bridges the gap between theory and practice by bringing forward a practically grounded and empirically validated framework for long-term management of surgery queues within complex healthcare settings.

2. Description of the case study

Hospital de Clínicas Unicamp (HC - Unicamp) is a university hospital located in Campinas, Brazil (HC - Unicamp, 2024b). It is a high-complexity public hospital serving approximately 6.5 million inhabitants from Campinas' metropolitan area and other regions of Brazil that handles around 500,000 patient cases per year. Of these, roughly 10,000 correspond to elective surgical procedures, with the remainder involving outpatient consultations and emergency care (HC - Unicamp, 2024a). In addition to providing healthcare services, HC - Unicamp operates as a teaching and research hospital, training healthcare professionals and conducting medical and interdisciplinary research.

Elective surgical patients follow a complex, multi-stage care pathway (see Figure 1), encompassing pre-operative, intra-operative and post-operative phases. This process begins with waiting-list management, initiated at local medical consultations and coordinated through the regional bed regulation centre (HC - Unicamp, 2024c), a government agency responsible for regulating hospital access across Campinas Metropolitan Region. Patient entry into HC - Unicamp typically occurs via scheduled outpatient appointments, although some emergency cases subsequently become elective and are incorporated into the same planning process. Once referred, patients are assigned to waiting lists according to medical subspecialty. Decisions at this stage directly determine the CMP (i.e., strategic layer), influencing waiting times, speciality-level demand balance and the long-term utilisation of operating theatre capacity (Siqueira et al., 2018).

On the day of surgery, patients enter the upstream unit management phase, where clinical reassessment is performed. Based on clinical requirements, Intensive Care Unit (ICU) support may be required pre- or postoperatively, necessitating prior bed reservation; otherwise, postoperative ward bed availability must be ensured. Patients then undergo pre-anaesthetic evaluation, diagnostic tests and surgical safety checks. If contraindications arise, the procedure is suspended and the patient is either discharged or rescheduled, according to clinical severity. If cleared, the patient

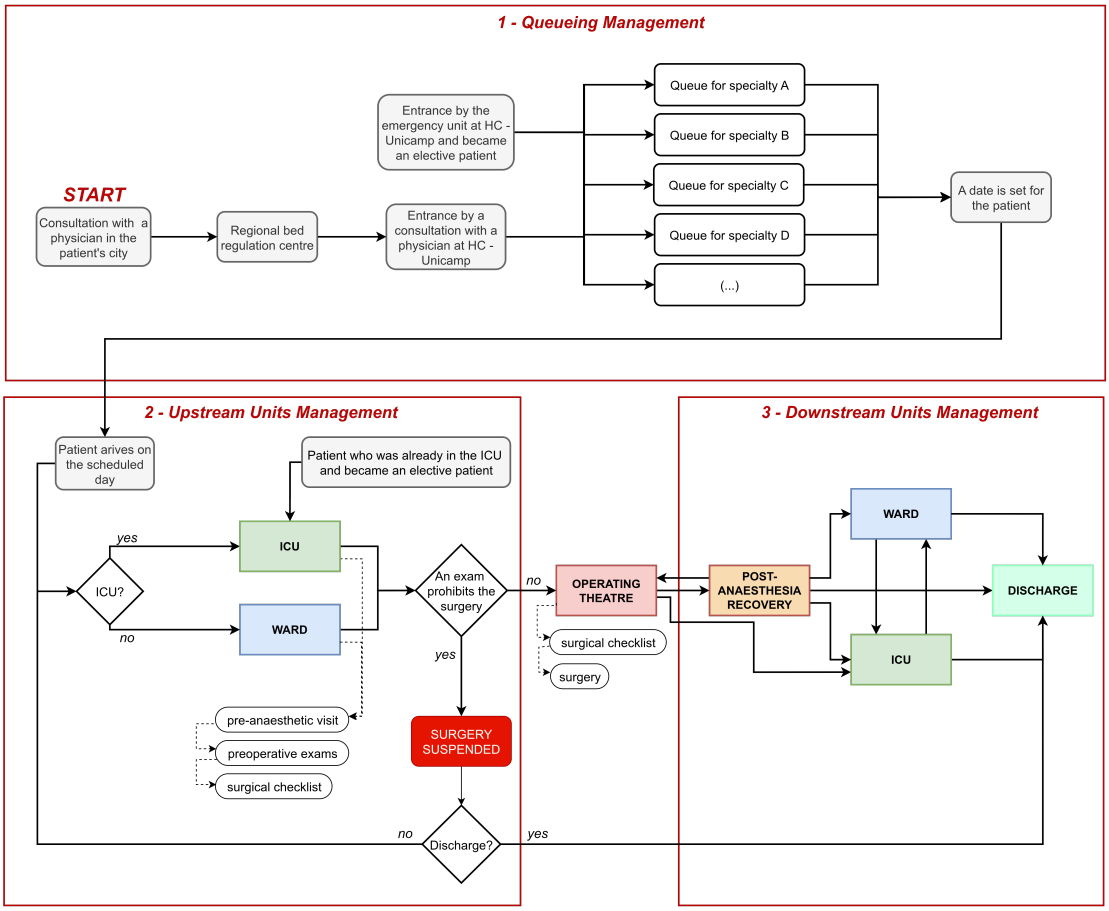


Figure 1: Patient flow at HC - Unicamp.

proceeds to the operating theatre, where the surgical procedure and subsequent theatre cleaning are performed. In the present study, ward and ICU capacity are sufficient and therefore do not constitute binding constraints, allowing the analysis to focus on waiting-list management, uncertainty in surgery duration and operating theatre allocation.

Once in the operating theatre, the patient is managed by a dedicated surgical team and anaesthetist. This stage includes final safety checks, the surgical procedure itself and post-operative cleaning. Here, uncertainty in surgery duration plays a central role, as it directly affects idle time, overtime, and the risk of cancellations. Indeed, it is well known that surgery duration variability has a significant impact on operating theatre utilisation and overall system throughput (e.g., Pandit and Tavare, 2011), therefore meriting explicit consideration in the proposed framework.

Following surgery, patients are transferred to the Post-Anaesthesia Recovery Unit (PARU), with direct ICU transfer occurring only in exceptional cases. After stabilisation, patients are admitted to a ward or ICU bed and subsequently discharged. HC - Unicamp has 15 operating theatres dedicated to elective surgeries: 13 located in the Elective Surgery Centre (ESC) and 2 in the Ambulatory Surgery Centre (ASC), which operates exclusively during morning shifts. These theatres are heterogeneous with respect to infrastructure and equipment, leading to compatibility restrictions across medical subspecialties.

The planning horizon comprises approximately 20 working days per month, each divided into two 6-hour shifts (morning and afternoon). Henceforth, we refer to a theatre shift as a *block*. Hospital policy assigns each block to a single subspecialty, while allowing the same subspecialty to occupy multiple blocks within the same theatre, provided compatibility conditions are met. Sequential allocation is therefore permitted (and preferred by hospital managers), enabling a subspecialty assigned to a morning block to also occupy the corresponding afternoon block.

Each month, operating theatres must be allocated across 40 medical subspecialties. Relevant subsets include: (i) those that mandatorily require two consecutive blocks in the same theatre due to the extended duration of their procedures; (ii) priority subspecialties, which take precedence in allocation decisions in accordance with institutional rules; and (iii) transplant subspecialties, which impose additional operational constraints.

Block allocation feasibility depends mainly on the availability of surgical teams and anaesthetists. A subspecialty can only be assigned to a block if an appropriate team is available on the corresponding day and shift. Moreover, the number of simultaneous allocations requiring anaesthesia cannot exceed the number of anaesthetists available in each period. For example, if 15 anaesthetists are available on a given morning, at most 15 anaesthesia-dependent procedures may be scheduled concurrently.

Day	PERIOD/THEATRE		1	2	3	4	5	6	7	8	9	10	11	12	17	ASC-1	ASC-2
1	MONDAY	MORNING	card sur	tumour orth		procto	uro onco	anaest dor		gast liv	otorhi			vasc			
		AFTERNOON	card sur	tumour orth	pc local	procto	uro onco				otorhi	gast gdp/vb		vasc	vasc local		
2	TUESDAY	MORNING	card sur	foot orth	plast	trauma orth	uro tx		uro tx	gast eed	otorhi	gast eed	neuro	trauma orth			
		AFTERNOON	card sur		plast						otorhi	gast eed	neuro	trauma orth			
3	WEDNESDAY	MORNING	card sur	knee orth	plast	thorax sur	vasc local	paed sur	plast		otorhi	head neck	neuro	trauma orth	vasc		
		AFTERNOON	card sur		plast	thorax sur		paed sur	plast		otorhi	gast obes	neuro		vasc		
4	THURSDAY	MORNING	card sur	eef	adult endo	gast gdp/vb	uro lithiasi	vasc	adult endo	gast liv	otorhi	gast obes	neuro	neuro colu		ophtal	
		AFTERNOON	card sur	pc local	ophtal	gast gdp/vb	uro lithiasi	vasc	otorhi	gast liv	otorhi	gast obes	neuro	neuro colu			
5	FRIDAY	MORNING		paed orth	plast		uro gyneco	paed sur			head neck	head neck	neuro colu			hand orth	
		AFTERNOON		hip orth										trauma orth			

Figure 2: Excerpt from a week's operating theatre schedule.

Figure 2 presents an excerpt from a monthly operating theatre schedule. Rows correspond to day and shift and columns to theatres; allocated blocks are colour-coded (blue for procedures requiring anaesthesia and grey otherwise). Unallocated (blank) slots reflect the complexity of the constraint set. Allocations shown in red correspond to transplant surgeries, which require the simultaneous allocation of two theatres within the same day and shift.

Historically, schedules such as those shown in Figure 2 have been manually constructed using only electronic spreadsheets, often requiring more than two full working days. This manual process is time-consuming and does not guarantee optimality with respect to institutional policies. Automating the scheduling process therefore reduces administrative burden and enables the generation of optimal solutions under complex constraints.

In addition to the impact of automating surgery scheduling in a reference hospital in Brazil, this paper provides an approach that is firmly grounded in reality and is therefore adaptable to other general hospitals. By jointly managing waiting-list dynamics, surgery duration uncertainty and operating theatre allocation, our proposed approach provides a coherent and practically grounded solution capable of satisfying waiting-time requirements while mitigating idle time, overtime and cancellation risks.

3. Proposed Framework

We propose an integrated framework for elective surgery planning under uncertainty that links waiting-list management, in-block time management, and operating theatre scheduling. Rather than formulating these decisions as a single monolithic model, the framework decomposes the problem into three inter-related modules. The first two modules address strategic decisions related to queue and overtime management, and their outputs are then used as inputs to the tactical operating theatre scheduling model. Figure 3 summarises the proposed framework.

The queue management module establishes targets for the number of surgeries to be performed at each planning period (e.g., monthly). Specifically, a modified (R, Q) policy defines, for each medical sub-speciality, a lower and an upper bound on the number of patients to be admitted for surgery during the planning period, while ensuring that system-wide waiting-time requirements are satisfied in spite of the uncertainty in patient arrivals.

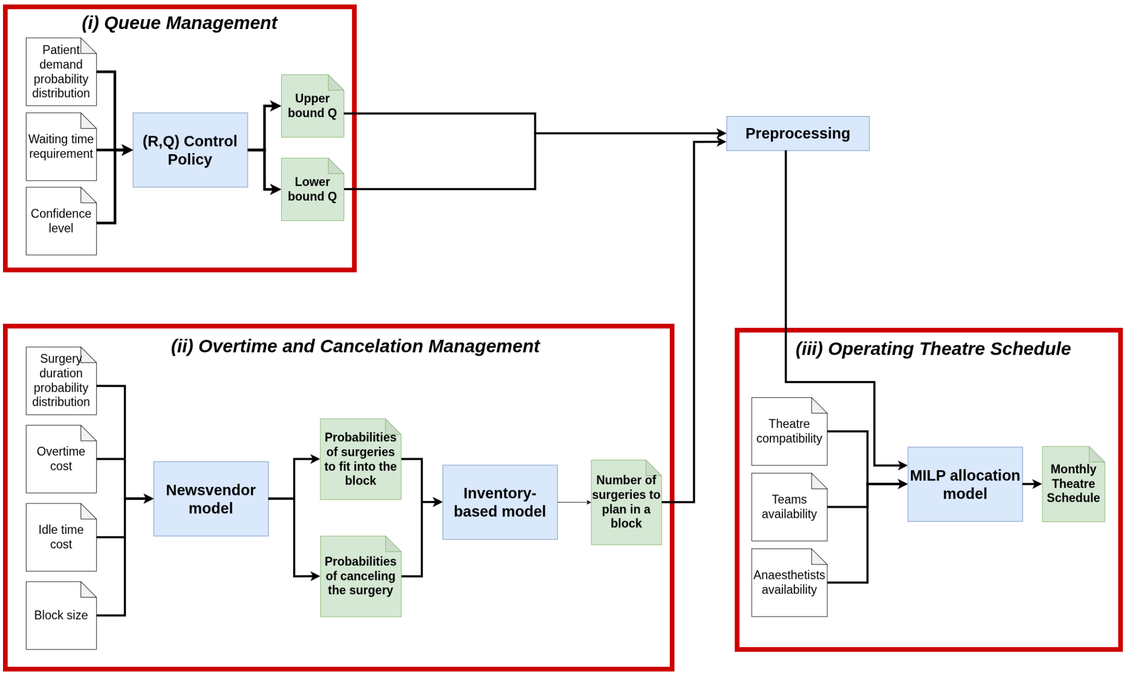


Figure 3: Proposed framework.

The second module balances overtime, idle time and the risk of surgery cancellation due to overbooking. Considering the uncertainty in surgery duration, a Newsvendor model guides dynamic decisions on whether to perform or postpone the next surgery, given the time remaining in the session. The output of the Newsvendor model then informs an inventory model that determines the number of surgeries to be booked in a block while balancing the expected number of surgeries with the risk of cancellation.

The outputs of the first two modules are then combined to provide key inputs to the third module, which formulates the OT scheduling problem. Specifically, given the probability distribution of the number of performed surgeries per session (module 2), and the bounds on the number of surgeries to be performed in the period (module 1), one can derive lower and upper limits on the number of blocks to be allocated to each surgical sub-speciality. Hence, theatre allocation decisions are informed not only by resource availability and clinical compatibility, but also include constraints to guarantee that waiting-time requirements can be met despite the uncertainties in the demand and duration of surgeries. Crucially, the scheduling model remains deterministic, thereby facilitating practical implementation and ensuring rapid solution times.

Overall, the proposed framework provides a coherent approach to elective surgery planning across multiple decision levels. By explicitly incorporating uncertainty into the strategic planning stages, it supports more robust and operationally meaningful scheduling decisions that are guaranteed to align to strategic waiting-time requirements. The following subsections present each module in detail. To help navigate the paper, Table 1 consolidates the list of acronyms.

3.1. Module 1: Long-Term Waiting List Management

CMP is the strategic decision layer that decides the volumes of elective surgeries to be performed over a given planning horizon, see Section 1. In contrast to much of the existing literature, which primarily selects these volumes to optimise financial or capacity-related objectives, this work adopts a stability-oriented perspective. We aim to enable decision-makers to impose maximum acceptable waiting times (e.g., six months) and to guarantee, in the long run, that such targets are met without allowing waiting lists to grow unboundedly. This is particularly relevant in the context of public healthcare systems, such as Brazil's Unified Health System (SUS), where there is a legal obligation to provide care to all patients in need. Thus, ensuring equitable access to care and adherence to waiting time standards is paramount.

Table 1: Acronyms used in the paper.

Acronym	Description
CMP	Case Mix Planning
MSS	Master Surgery Scheduling
SCS	Surgical Case Scheduling
OT	Operating Theatre
MDP	Markov Decision Process
SUS	Brazilian Unified Health System
HC - Unicamp	Hospital de Clínicas Unicamp
OTSP	Operating Theatre Scheduling Problem
MILP	Mixed-Integer Linear Programming
PMF	Probability Mass Function
CDF	Cumulative Distribution Function

In line with this objective, this section introduces the theoretical foundations of the modified (R, Q) control policy proposed to manage elective surgery waiting lists under stochastic demand. We first characterise the dynamics of the underlying stochastic process and analyse its long-run behaviour. Subsequently, we derive analytical upper and lower bounds on the batch size Q that ensures compliance with prescribed waiting-time requirements, thereby establishing the strategic input for the integrated planning framework. Table 2 summarises the notation used in this section.

3.1.1. (R, Q) *surgery scheduling policy: long-term distributions*

Let X_t, $t \geq 0$ be a stochastic process representing an elective surgery queue with stationary and independent increments over discrete planning periods, with X_t representing the number of patients in the system at period $t \geq 0$. Assume that the queue is controlled via a modified (R, Q) policy (e.g., Axsäter, 2015) whereby a batch of $Q \in \mathbb{N}$ surgeries is processed at time period t whenever $X_t \geq R$, $R \geq Q$ at the start of the period, where $\mathbb{N}$ is the set of positive integers. In contrast to the classical (R, Q) policy, which processes as many batches as necessary to bring the inventory below the reorder point (R), our policy can only process a batch at a time and relies on long-term properties to ensure that the process X_t, $t \geq 0$ is stable and guarantees prescribed waiting time requirements.

Let D be a non-negative integer random variable representing the overall demand between two consecutive planning periods, with probability distribution $f_D : \Omega \to [0, 1)$, $\Omega \subset \mathbb{Z}_+$, and cumulative distribution $F_D : \mathbb{Z}_+ \to [0, 1]$ given by

$$F_D(d) = \sum_{l=0}^{d} f_D(l),$$

where $\mathbb{Z}_+$ denotes the set of non-negative integers. This implies that the dynamics of process X_t, $t \geq 0$ can be expressed as:

$$X_{t+1} = X_t + d - Q\mathbb{1}_{\{X_t \geq R\}}, \tag{1}$$

where $\mathbb{1}_{\{A\}}$ is the indicator function, which is equal to 1 if A holds and is nil otherwise, and d is a value drawn from f_D that represents the surgical demand in period t. The state space of process X_t, $t \geq 0$ is given by $S = \{0, 1, 2, \ldots\}$, where $X_t = i$ indicates that there are $R - Q + i$ patients waiting for elective surgery.

Letting

$$Y_t = \left\lfloor \frac{X_t}{Q} \right\rfloor, \qquad Z_t = X_t \,\%\, Q, \tag{2}$$

where % represents the modulus operator, we can write:

$$X_t = QY_t + Z_t,$$

where Y_t, $t \geq 0$ is the integer number of batches required to bring the queue instantaneously below R and $Z_t \in \{0, 1, \ldots, Q - 1\}$ is the number of patients that would remain in excess of the minimum allowed queue of $(R - Q)$ patients after removing Y_t batches of Q patients instantaneously.

Note that, from the definitions above, the state space of process Y_t, $t \geq 0$ is $S_Y = \mathbb{Z}_+$, while $S_Z = \{0, 1, \ldots, Q-1\}$ is the state space of process Z_t, $t \geq 0$. In the next result, we will derive the long-term distribution of process Z_t, $t \geq 0$.

Table 2: Symbols used in the long-term waiting list management model.

Index	Description
t	Planning period index
z	State index of the remainder process
i, j	Generic state indices
Set	**Description**
S	State space of process X_t
S_Y	State space of process Y_t
S_Z	State space of process Z_t
Ω	Support of the demand random variable
Parameter	**Description**
R	Reorder threshold of the modified (R, Q) policy
Q	Batch size under the modified (R, Q) policy
τ	Waiting-time target
α	Probability threshold for the waiting-time requirement
$\overline{Q}$	Upper bound on feasible batch sizes
$\underline{Q}$	Lower bound on feasible batch sizes
$\overline{M}$	Maximum number of batch arrivals in one period
Random Variable / Stochastic Process	**Description**
X_t	Queue-length process
D	Demand between two consecutive planning periods
d	Realisation of D
Y_t	Batch-level queue process
Z_t	Remainder process associated with X_t
A	Number of batch arrivals in one period
ω	Waiting time of an arriving patient
Function	**Description**
$f_D(\cdot)$	Probability mass function of D
$F_D(\cdot)$	Cumulative distribution function of D
$A(z)$	Number of batch arrivals given remainder state z
$\pi_Z(\cdot)$	Steady-state distribution of Z_t
$\pi_Y(\cdot)$	Steady-state distribution of Y_t
P_Z	Transition matrix of process Z_t
P^Y	Transition matrix of process Y_t
$p^Z_{zz'}(k)$	Conditional transition probability of Z_t given demand k
p_{ij}	Generic transition probability entry
Derived Quantity	**Description**
$\mathbb{1}_{\{X_t \geq R\}}$	Indicator of batch processing

Proposition 1. *Let $\pi_Z : S_Z \to [0, 1]$ be the steady state distribution of process X_t, $t \geq 0$. Then, it holds that*

$$\pi_Z(z) = \frac{1}{Q}, \forall z \in S_Z. \tag{3}$$

Proof. The proof closely follows that of Axsäter (2015, Proposition 5.1). Let P_Z be the transition matrix describing the dynamics of process Z_t, $t \geq 0$. Assume that $Z_t = z$, $z \in S_Z$ at the start of period $t \geq 0$, and let and $D = k$ be the number of new patient arrivals in period t. Hence, it follows that

$$Z_{t+1} = (z + k) \,\%\, Q = (z + k + QY_t) \,\%\, Q = (X_t + k) \,\%\, Q,$$

where the last equality come from the definitions in Eq. (2). Therefore, given $D = k$, it is evident that $p^Z_{zz'}(k) := P(Z_{t+1} = z' | Z_t = z, D = k)$ equals one for exactly one value of z and zero otherwise. Consequently, we can write:

$$\sum_{z=0}^{Q-1} p^Z_{zz'} = \sum_{z=0}^{Q-1} \mathbb{E}_k \left\{ p^Z_{zz'}(k) \right\} = \mathbb{E}_k \left\{ \sum_{z=0}^{Q-1} p^Z_{zz'}(k) \right\} = \mathbb{E}_k\{1\} = 1.$$

The expression above implies that the Markov chain describing the dynamics of process Z_t, $t \geq 0$ is doubly stochastic, and therefore has a uniform steady state distribution. Indeed, substituting (3) in the steady state equations:

$$\pi^Z(z) = \sum_{z'=0}^{Q-1} \pi^Z(z') p^Z_{z'z},$$

one can easily see that the steady state distribution satisfies Eq. (3). □

The result in Proposition 1 is very important, as it shows that the steady state distribution of process Z_t, $t \geq 0$, is uniform regardless of the distribution of patient arrivals. Next, we will use this fact to derive the steady state distribution of process Y_t, $t \geq 0$.

3.1.2. The dynamics of process Y_t, $t \geq 0$

Assume that process Z_t, $t \geq 0$ has reached equilibrium, which is also equivalent to supposing that $P(Z_0 = z) = \pi_Z(z) = \frac{1}{Q}$ (e.g., Brémaud, 2020). We can describe the dynamics of process Y_t, $t \geq 0$ as:

$$Y_{t+1} = \begin{cases} Y_t - 1 + A, & \text{if } Y_t \geq 0, \\ A; & \text{if } Y_t = 0, \end{cases} \tag{4}$$

where A is the round number of batch arrivals at time t. To fully characterize the random variable A, it is necessary to define

$$A(z) = \left\lfloor \frac{D+z}{Q} \right\rfloor, \tag{5}$$

which represents the round number of batch arrivals at time t, given that $Z_t = z$. Then, in view of Proposition 1, it follows that:

$$P(A = k) = \frac{1}{Q} \sum_{z=0}^{Q-1} P(A(z) = k). \tag{6}$$

Therefore, from (4) we have:

$$p^Y_{ij} = P(Y_{t+1} = j | Y_t = i) = \begin{cases} P(A = j), & \text{if } Y_t = 0, \\ P(A = j - i + 1), & \text{if } Y_t \geq 1. \end{cases} \tag{7}$$

Note that Eq. (4)-(7) correspond to the dynamics in Shortle et al. (2018, Section 6.1.2). Similarly to Shortle et al. (2018), the transition matrix of process Y_t, $t \geq 0$ can be written as:

$$P^Y = [p_{ij}] = \begin{bmatrix} k_0 & k_1 & k_2 & \dots & k_M & 0 & 0 & \dots \\ k_0 & k_1 & k_2 & \dots & k_M & 0 & 0 & \dots \\ 0 & k_0 & k_1 & k_2 & \dots & k_M & 0 & \dots \\ 0 & 0 & k_0 & k_1 & k_2 & \dots & k_M & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \end{bmatrix}, \tag{8}$$

where M is the maximum number of new batch arrivals, i.e., the maximum value that random variable A can assume.

The steady state probabilities of process Y_t, $t \geq 0$ can then be recursively evaluated by solving:

$$\pi_Y^T = \pi_Y^T P, \tag{9}$$

with the contour condition that $\pi_Y(0) = 1 - \rho$ (Shortle et al., 2018, Eq. (6.12)), where $\rho = E(A)$. Observe that, to guarantee stability, we must have $E(A) < 1$. Note also that the steady state probabilities up to a given state j can be recursively calculated using the first j equations in (9).

3.1.3. *Upper and lower bounds on the batch size Q under waiting time requirements*

Let ω be the waiting time of a patient joining the queue, and assume that the healthcare system requires that:

$$P(\omega > \tau) < \alpha, \tag{10}$$

for some target waiting time $\tau > 0$ and probability threshold $\alpha \in (0, 1)$. Now, if a patient arrives and encounters $Y_t > 0$ batches of patients in the queue, their turn will only arrive once these Y_t batches are processed, which will take Y_t time periods. This means that the patient will have to wait $\tau = Y_t$ periods for their turn. Therefore, Eq. (10) can be equivalently stated as:

$$\sum_{i=0}^{\tau} \pi_Y(i) > 1 - \alpha, \tag{11}$$

where $\pi_Y : S_Y \to (0, 1)$ satisfies Eq. (9). Therefore, to limit the waiting time according to the prescribed requirements, we can establish a lower bound ($\underline{Q}$) on the batch size $Q > 0$ as follows:

$$\underline{Q} = \min\{Q \in \mathbb{N} : \sum_{i=0}^{\tau} \pi_Y(i) > 1 - \alpha\}. \tag{12}$$

When $Y_t = 0$, on the other hand, the next batch will only be processed when the system leaves this state, therefore by applying the strong Markov property (Brémaud, 2020), we have:

$$\omega = \min\{\delta > 0 : Y_\delta > 0 | Y_0 = 0\}.$$

Considering the transition matrix in Eq. (7), it follows that ω will be a geometrically distributed variable and:

$$P(\omega > \tau | Y_0 = 0) = k_0^\tau. \tag{13}$$

One can intuitively see that the value of k_0 will increase as Q increases, since in this case we will need more patient arrivals until we are able to process a full batch of patients. Hence, from Eq. (13), we can set up an upper bound $\overline{Q}$ for the batch size such that:

$$\overline{Q} = \max\{Q \in \mathbb{N} : k_0^\tau < \alpha\}. \tag{14}$$

In the example below, we show how to derive general bounds $\underline{Q}$ and $\overline{Q}$ for a given demand distribution and waiting time requirements.

Example 1 (Deriving lower and upper bounds for Q)**.** *Consider an elective surgery queue with monthly demand D that satisfies the distribution in Table 3.*

Table 3: Monthly demand distribution for surgeries

Demand (d)	0	1	2	3	4	5	6	7	8	9	10
$P(D = d)$	0.0308	0.0393	0.1179	0.2096	0.2445	0.1956	0.1086	0.0414	0.0103	0.0015	0.0005

The expected monthly demand is therefore $\mathbb{E}(D) = 3.9022$. Setting $Q = 4$, which ensures long-term stability, we have $S_Z = \{0, 1, 2, 3\}$. This allows us to evaluate the individual distributions of $A(z)$, $z \in S_Z$ as defined in Eq. (5)*, which we depict in the Table 4, along with the distribution of variable A from Eq.* (6)*.*

The distributions in Table 4 follow directly from Eq. (5) *and Eq.* (6)*. For example, when $z = 0$, we know that the queue comprises an integer number of batches. Therefore, $P(A(0) = 0) = p_D(0) + p_D(1) + p_D(2) + p_D(3)$ (Table 3) as up to 3 new arrivals will not suffice to generate a new batch of $Q = 4$ patients. Analogously, $A(0) = 1$ if we observe from 5 to 7 arrivals and $A(0) = 2$ when 8 to 10 patients arrive in the period, as these will suffice for 2 new batches.*

Conversely, when $z = 3$, we have three patients in excess of a discrete number of batches, which means that $P(A(3) = 0) = p_D(0)$ as new batches will be formed whenever one or more patients arrive. One new batch will

Table 4: Monthly demand distribution for surgeries

$P(A(z) = d)$	**Batch demand** (d)			
	0	1	2	3
$z = 0$	0.3976	0.5901	0.0123	0
$z = 1$	0.1880	0.7583	0.0537	0
$z = 2$	0.0701	0.7676	0.1618	0.0005
$z = 3$	0.0308	0.6113	0.3559	0.002
$P(A = d)$	**0.171625**	**0.681825**	**0.145925**	**0.000625**

form if $D \in [1, 4]$, two new batches if $D \in [5, 8]$ and three new batches when $D \in [8, 10]$. Specifically, when 10 patients arrive, they will form a set of 13 patients when joined to the $z = 3$ remaining patients from the start of the period. These 3 patients will then suffice to form three batches of $Q = 4$ patients, with one remaining batchless for the following period.

Note that $\mathbb{E}(A) = 0.9755$, and that this value is equal to $\rho = \frac{\mathbb{E}(D)}{Q} = \frac{3.9022}{4} = 0.9755$. This is expected, as the dynamics of process Y_t describes the original system, only looking at it in terms of batches of patients. Note that $\mathbb{E}(A)$ is simply the expected number of batches of Q patients arriving and since we process $\mu = 1$ batch per time period, we have $\rho = \mathbb{E}(A)$ when looking at the evolution of the batch process Y_t, $t \geq 0$.

Now, let us suppose that $\tau = 6$ and $\alpha = 0.05$ in (10), which means that the system requires 95% of the patients to be operated in up to six months. To verify whether that is satisfied, we need firstly to solve Eq. (9) for the steady state distribution of Y_t, $t \geq 0$ and then use it to verify (11). From Table 4, we obtain $k_0 = 0.171625$, $k_1 = 0.681825$, $k_2 = 0.145925$ and $k_3 = 0.000625$, which can be applied in Eq. (9) to find $\pi_Y(0) = 0.02445$, $\pi_Y(1) = 0.1182$, $\pi_Y(2) = 0.1219$, $\pi_Y(3) = 0.1046$, $\pi_Y(4) = 0.0898$, $\pi_Y(5) = 0.077$, $\pi_Y(6) = 0.0661$. This yields:

$$P(\omega \leq 6 | Y_t \geq 1) \approx 0.602 < 0.95 = 1 - \alpha.$$

This implies that $\underline{Q} > 4$ in Eq. (12) and we should, therefore, recalculate the probability above for increasing values of Q until we find the first value that satisfies the waiting time criterion - which will give us the lower bound $\underline{Q}$.

To verify the waiting time requirements when $Y_t = 0$, one can apply (13), which implies that $P(\omega \leq 6 | Y_t = 0) = 0.171625^6 \approx 2.5 \cdot 10^{-5} < 0.05$. Therefore, to find the bound $\overline{Q}$ in Eq. (14), we must iterate on higher values of Q.

When verifying Eq. (12) with $Q = 10$, it implies $P(\omega \leq 6 | Y_t = 0) = 0.609780^6 \approx 0.05114 > 0.05$. Then, if one set $Q = 9$, it implies $P(\omega \leq 6 | Y_t = 0) = 0.566478^6 \approx 0.03304 < 0.05$. Thus, we can set $\overline{Q} = 9$.

For the $\underline{Q}$, one can verify $Q = 5$, which implies $k_0 = 0.265720$, $k_1 = 0.688120$, $k_2 = 0.046160$ and $k_3 = 0$, which can be applied in Eq. (9) to find $\pi_Y(0) = 0.21956$, $\pi_Y(1) = 0.606723$, $\pi_Y(2) = 0.143539$, $\pi_Y(3) = 0.024935$, $\pi_Y(4) = 0.004332$, $\pi_Y(5) = 0.000752$, $\pi_Y(6) = 0.000131$. This yields:

$$P(\omega \leq 6 | Y_t \geq 1) \approx 0.999973 > 0.95 = 1 - \alpha.$$

Thus, we can set $\underline{Q} = 5$.

3.2. Module 2: Modelling Uncertainty in Surgery Duration and Block Sessions

This section details the components of module 2 of the proposed framework, see Section 3. Firstly, it is necessary to estimate how many elective patients can be accommodated within a single operating theatre block for each sub-speciality, to ensure consistency between the periodic schedule and the waiting-time requirements established by the (R, Q) policy in Section 3.1. This is key to ensuring that the number of sessions allocated to a given speciality is enough to warrant a number of surgeries that lies in the interval $[\underline{Q}, \overline{Q}]$, considering the probabilistic distribution of surgery durations. Specifically, a trade-off arises between idle time, that results in underutilised theatre capacity, and overtime, which may trigger cancellations of scheduled procedures.

At first, we introduce a Newsvendor model to guide operational decisions on whether to perform or cancel the next scheduled surgery, depending on the time remaining in the session. Then, assuming that these decisions are taken sequentially at the end of each scheduled surgery, we introduce an inventory model to find the number of patients that can be reliably scheduled within a time block, as well as the probability distribution of the number of cancellations at each session. These are tactical inputs required for subsequent operating theatre allocation decisions.

3.2.1. *Managing surgery cancellations via a Newsvendor-based model*

Let us now consider the problem of determining the optimal time budget for the next surgery in a session, based on the random duration of this surgery. The symbols used in this section are defined in Table 5.

Table 5: Symbols used in the Newsvendor-based model.

Symbol	**Description**
Index	**Description**
s	Specialty index
Set	**Description**
$\mathcal{S}$	Set of specialties
Ω	Support of the surgery-duration random variable
Parameter	**Description**
T	Length of an operating theatre block
t_b	Time budget allocated to the last surgery in a block
c	Unit cost of unused allocated time
p	Unit penalty cost of overtime
t^*	Safety time threshold for starting the next surgery
Random Variable	**Description**
T_D	Random duration of a surgery
β_1	Unused allocated time
β_2	Overtime duration
Function	**Description**
$f_{T_D}(\cdot)$	Probability mass function of T_D
$F_{T_D}(\cdot)$	Cumulative distribution function of T_D
$u(t_b)$	Expected cost of allocating time budget t_b

Let T_D be a non-negative random variable taking values from $\Omega \in \mathbb{R}_+$, which represents the duration of a given surgery, with cumulative distribution $F_{T_D} : \Omega \to [0, 1]$. Assume that, once the theatre is ready for the surgery, it only proceeds as planned if the remaining time in the session exceeds a prescribed budget of $t_b \in \Omega$ time units. Otherwise, the surgery is called off and the session is terminated.

The time budget $t_b \in \Omega$ can be interpreted as a perishable inventory of session time. If the inventory is excessive, we will not fully utilise the time allocated to the surgical team to perform the surgery. Conversely, if we proceed with the surgery but the time budget is insufficient, the surgical team will have to work overtime to complete the surgery. Let $c > 0$ be the cost of each unit of time budget in excess of the actual surgical time T_D and $p > c$ be a penalty for each unit of overtime that the surgical team incurs. The expected cost of the time deviation between T_D and t_b is defined as:

$$u(t_b) = c\mathbb{E}(\beta_1) + p\mathbb{E}(\beta_2), \tag{15}$$

where $\beta_1 = \min\{t_b - T_D,\ 0\}$ is the underused time budget, and $\beta_2 = \max\{T_D - t_b,\ 0\}$ is the total overtime. The objective is to find an optimal time budget allocation t^* that minimizes the time deviation cost in Eq. (15), which is given by:

$$t^* = \arg\min_{t_b \in \Omega} u(t_b). \tag{16}$$

Lemma 1. *Let $\beta_1 = \min\{t_b - T_D,\ 0\}$ and $\beta_2 = \max\{T_D - t_b,\ 0\}$. Then, it follows that:*

$$\mathbb{E}(\beta_1) = \int_0^{t_b} F_{T_D}(t)dt, \quad \mathbb{E}(\beta_2) = \int_{t_b}^{\infty} (1 - F_{T_D}(t))dt \tag{17}$$

Proof. We will start by proving the result for β_1. Since $\beta_1 \geq 0$, it follows that:

$$\mathbb{E}(\beta_1) = \int_{t=0}^{\infty} P(\beta_1 > t)dt = \int_{t=0}^{t_b} P(t_b - T_D > t)dt = \int_{t=0}^{t_b} P(T_D < t_b - t)dt = \int_{t=0}^{t_b} F_{T_D}(t_b - t)dt = \int_{t=0}^{t_b} F_{T_D}(t)dt.$$

Since β_2 is also non-negative, we can write:

$$\mathbb{E}(\beta_2) = \int_0^{\infty} P(T_D - t_b > t)dt = \int_0^{\infty} P(T_D > t + t_b)dt = \int_{t_b}^{\infty} P(T_D > t)dt = \int_{t_b}^{\infty} (1 - F_{T_D}(t))dt$$

□

Theorem 1. *Let $u(t_b)$ be defined in Eq.* (15)*, and t^* be a solution to Eq.* (16)*. Then, if follows that:*

$$t^* = F_{T_D}^{-1}\left(\frac{\frac{p}{c}}{1+\frac{p}{c}}\right), \tag{18}$$

where $F_{T_D}^{-1}(\cdot)$ is the inverse of the cumulative distribution function $F_{T_D}(\cdot)$.

Proof. Substituting (17) in Eq. (15) yields:

$$u(t_b) = c\int_0^{t_b} F_{T_D}(t)dt + p\int_{t_b}^{\infty}(1 - F_{T_D}(t))dt.$$

When $t_b = t^*$, the first order optimality condition for $u(t_b)$ implies:

$$\frac{du(t_b)}{dt_b} = cF_{T_D}(t^*) - p(1 - F_{T_D}(t^*)) = 0.$$

Hence, we must have:

$$cF_{T_D}(t^*) - p(1 - F_{T_D}(t^*)) \implies \frac{F_{T_D}(t^*)}{(1 - F_{T_D}(t^*))} = \frac{p}{c} \implies F_{T_D}(t^*)\left(1 + \frac{p}{c}\right) = \frac{p}{c}$$

We conclude the proof by noting that (18) follows from the expression above. □

Theorem 1 implies that, whenever the remaining time in a session exceeds the quantile t^* in Eq. (18), the surgery is greenlighted to continue. Otherwise, the surgery is cancelled and the session comes to a close. Note that the quantile is solely a function of the ratio between the overtime cost p and the cost of unused session time c, and is applicable to any surgery time distribution. For example, if $p = 2c$, $t^* = F_{T_D}^{-1}\left(\frac{2}{3}\right)$. This means that at least two thirds of the green-lighted surgeries will not require overtime.

3.2.2. Applying the newsvendor model to the case study

Building on the theoretical results of the previous subsection, we first estimate, for each surgical subspecialty, the distribution of the number of surgeries that can be completed within a block of length $T \in \mathbb{Z}_+$. This distribution is then used in the subsequent subsection to evaluate session-planning decisions under an inventory-based formulation. For this purpose, we use the notation introduced in Table 5.

For each subspecialty, let d_i denote the non-negative random variable representing the duration of the i-th surgery, with probability mass function f_d and cumulative distribution function F_d. We assume that surgery durations are independent and identically distributed within each subspecialty. Hence, for any $k \in \mathbb{N}_+$, the cumulative duration of the first k surgeries is

$$cd_k = \sum_{i=1}^{k} d_i.$$

Let f_{cd_k} and F_{cd_k} denote the probability mass function and cumulative distribution function of cd_k, respectively.

After k surgeries have been completed, one further surgery can be accommodated only if the residual time in the block exceeds the safety threshold t^*. herefore,

$$P(K > k) = P(T - cd_k > t^*) = P(cd_k < T - t^*). \tag{19}$$

Thus, the event $\{K > k\}$ is equivalent to the event that, after completing k surgeries, sufficient time remains in the block to start one additional surgery while preserving the required safety buffer.

Table 6: Symbols used in the derivation of the completion distribution.

Set / Index / Parameter	Description
$\mathcal{S}$	Set of surgical subspecialties
s	Index of a surgical subspecialty, $s \in \mathcal{S}$
T	Length of the operating block
t^*	Safety time threshold that must remain available in the block to accommodate one further surgery
$\varepsilon_{\text{stop}}$	Probability threshold used to truncate the tail of the distribution
$K_{\max}$	Largest value of k such that $P(K > k) > \varepsilon_{\text{stop}}$
Random Variable	**Description**
d_i	Duration of the i-th surgery for a given subspecialty
cd_k	Cumulative duration of the first k surgeries, $cd_k = \sum_{i=1}^{k} d_i$
K	Number of surgeries that can be completed within a block
Function / Derived Quantity	**Description**
f_d	Probability mass function of the duration of a single surgery
F_d	Cumulative distribution function of the duration of a single surgery
f_{cd_k}	Probability mass function of cd_k
F_{cd_k}	Cumulative distribution function of cd_k
f_K	Probability mass function of K
F_K	Cumulative distribution function of K
$*$	Convolution operator
q	Vector with entries $q[k] = P(K > k)$

Because cd_k is the sum of k independent surgery durations, its distribution can be obtained recursively by convolution:

$$f_{cd_1} = f_d, \qquad f_{cd_k} = f_{cd_{k-1}} * f_d, \quad k \geq 2.$$

For each k, Eq. (19) then yields the survival probability $P(K > k)$.

These survival probabilities are computed sequentially in Algorithm 1 and stored in the vector q, whose k-th entry is defined as

$$q[k] = P(K > k).$$

The procedure starts at $k = 0$, with $q[0] = 1$, and stops when $q[k] \leq \varepsilon_{\text{stop}}$. This truncation preserves the relevant support of the distribution while avoiding unnecessary convolution steps in the negligible tail. For each subspecialty, the algorithm therefore returns both the vector q and the largest relevant index k, defined as $K_{\max}$.

Algorithm 1 Computation of $P(K > k)$ via convolutions

Input: T, t^*, $\varepsilon_{\text{stop}}$, subspecialties $s \in \mathcal{S}$, and duration distributions f_d for each $s \in \mathcal{S}$
Output: vector q such that $q[k] = P(K > k)$
for each $s \in \mathcal{S}$ **do**
  Initialise $q[0] \leftarrow 1$
  Initialise $k \leftarrow 1$
  Initialise $f_{cd_1} \leftarrow f_d$
  while $q[k-1] > \varepsilon_{\text{stop}}$ **do**
    if $k > 1$ **then**
      $f_{cd_k} \leftarrow f_{cd_{k-1}} * f_d$
    end if
    $q[k] \leftarrow P(cd_k < T - t^*)$
    $k \leftarrow k + 1$
  end while
  $K_{\max} \leftarrow k - 1$
end for

Once the survival function has been obtained, the probability mass function of K follows from the standard relationship between the PMF and the survival function of a discrete random variable:

$$P(K = k) = P(K > k - 1) - P(K > k), \qquad k \geq 1. \tag{20}$$

In addition,

$$P(K = 0) = 1 - P(K > 0).$$

Hence, the convolution procedure yields the full probability distribution of the number of surgeries that can be completed within the block for each subspecialty. This distribution is the main output of the present subsection and constitutes the probabilistic input to the inventory-based planning model described next.

3.2.3. Inventory-based model for defining the number of surgeries in a time block

Using the distribution of K derived in the previous subsection, we now determine the number of surgeries to be planned within a single operating theatre block. This decision is formulated through an inventory-based perspective that balances the marginal benefit of scheduling additional surgeries against the expected cost of cancellations (e.g., Axsäter, 2015). The symbols used in this subsection are defined in Table 7.

Table 7: Symbols used in the inventory-based planning model.

Parameter / Decision Variable	**Description**
n	Number of surgeries planned in a block
n^*	Optimal number of surgeries planned in a block
ℓ	Number of cancelled surgeries
g	Marginal value associated with scheduling one additional surgery
Function / Derived Quantity	**Description**
$G(n)$	Marginal gain associated with planning n surgeries
$cost(\ell)$	Cost incurred when ℓ surgeries are cancelled
$C(n)$	Expected cancellation cost when n surgeries are planned
$\Delta(n)$	Net value of planning n surgeries
$P(\text{cancellations} = \ell \mid n)$	Probability of cancelling ℓ surgeries when n surgeries are planned

Let $n \in \mathbb{N}_+$ denote the number of surgeries planned in a block. If the random number of surgeries that can actually be completed is K, then the number of cancellations is given by

$$\ell = (n - K)^+,$$

where $(x)^+ = \max\{x, 0\}$. Hence, for $\ell = 0, 1, \ldots, n$,

$$P(\text{cancellations} = \ell) = P(K = n - \ell), \qquad \ell = 1, \ldots, n, \tag{21}$$

and

$$P(\text{cancellations} = 0) = P(K \geq n).$$

We next associate each planning decision with a marginal gain and an expected cancellation cost. In contrast to classical inventory formulations, in which the reward is usually proportional to the expected number of units served, we define the gain of adding an $n-$th surgery as the value of successfully accommodating the n-th scheduled surgery:

$$G(n) = g \cdot \sum_{j=0}^{n} P(K \geq j), \tag{22}$$

where $g > 0$ is a constant representing the marginal value of scheduling one additional surgery. Since

$$P(K \geq j) = P(K > j - 1),$$

Eq. (22) is directly obtained from the survival probabilities computed in the previous subsection.

In this setting, $G(n)$ is not interpreted as a direct monetary return, but rather as the marginal benefit associated with increasing the number of planned surgeries from $n - 1$ to n. This interpretation is particularly suitable in public healthcare settings, where the value of additional surgeries may reflect improved access to care, reduced waiting times, and better utilisation of operating theatre capacity.

The cost associated with cancellations is modelled through a function $cost : \mathbb{Z}_+ \to \mathbb{R}_+$ satisfying $cost(0) = 0$ and assumed to be non-decreasing. The expected cancellation cost under planning decision n is therefore

$$C(n) = \mathbb{E}[cost((n - K)^+)] = \sum_{x=0}^{n} P(K = x)\, cost((n - x)^+). \tag{23}$$

The net value associated with planning n surgeries is then defined as

$$\Delta(n) = G(n) - C(n). \tag{24}$$

Because $G(n)$ represents the overall benefit scheduling n surgeries, the optimal planning level is the largest value for which the marginal net value remains positive:

$$n^* = \max\{n \in \mathbb{N}_+ : \Delta(n) > 0\}. \tag{25}$$

Thus, n^* identifies the point at which the expected benefit of adding one more planned surgery is no longer sufficient to offset the corresponding expected cancellation cost.

Once n^* has been determined, the relevant probabilistic input for the subsequent operating theatre allocation model is the truncated distribution of the number of surgeries completed under that planning decision, i.e.,

$$P(K = k), \qquad k = 0, 1, \ldots, n^*,$$

together with the implied probability distribution of the number of cancellations. These quantities are then passed to the next stage of the framework.

3.3. Operating Theatre Scheduling Problem

Following the proposed integrated framework, the final step consists in solving the Operating Theatre Scheduling Problem (OTSP), which addresses the tactical allocation of surgical subspecialties to operating theatre blocks. This stage translates the strategic decisions derived in the preceding modules into implementable schedules for hospital operations. In particular, the OTSP takes as input the target number of surgeries for each sub-speciality, as determined by the (R, Q) policy, together with the effective block capacity obtained from the cancellation and overtime management module.

We formulate the OTSP as a Mixed-Integer Linear Programming (MILP) model. This formulation allows the explicit representation of operational constraints and multiple, potentially competing objectives, while ensuring optimality with respect to the specified criteria. The model is designed to reflect hospital priorities such as high operating theatre utilisation, the allocation of priority subspecialities, and the efficient sequencing of blocks from a logistical perspective.

The proposed framework is, however, not restricted to the MILP approach adopted here. Since the previous modules provide demand and effective capacity as structured inputs, alternative solution methods, including heuristics and metaheuristics, could also be employed in the OTSP stage. The main value of the framework therefore lies in the integration of decision layers, rather than in the exclusive use of a particular scheduling technique.

In the following subsections, we first describe how the strategic outputs of the (R, Q) policy and the effective capacity information are preprocessed into block-demand units, which constitute the relevant unit of the scheduling problem. We then present the MILP formulation of the OTSP, including its decision variables, objective functions, and constraints.

3.3.1. Preprocessing Strategic Input

The OTSP is formulated at the block level. Therefore, the strategic outputs of the previous modules must be converted into the number of blocks required for each subspecialty.

For each subspecialty $s \in \mathcal{S}$, let

$$p_s(k) = P(K = k), \qquad k = 0, 1, \ldots, n_s^*,$$

denote the probability mass function of the number of surgeries that can be completed in a single block under the optimal planning decision obtained in the cancellation and overtime management module. This distribution summarises the effective stochastic capacity of one block for subspecialty s.

To determine the number of blocks required to achieve activity targets $\underline{Q}$ and $\overline{Q}$ at confidence level α, we use successive convolutions of $p_s(\cdot)$. Let

$$p_s^{(n)} = \underbrace{p_s * \cdots * p_s}_{n \text{ times}}$$

denote the probability mass function of the total number of surgeries that can be completed across n blocks allocated to subspecialty s, and let $F_s^{(n)}$ be its corresponding cumulative distribution function.

Then, for a generic target $Q' \in \{\underline{Q}, \overline{Q}\}$, the minimum number of blocks required is the smallest integer $n \in \mathbb{N}_+$ such that

$$P\left(\sum_{b=1}^{n} K_b \geq Q'\right) = 1 - F_s^{(n)}(Q' - 1) \geq \alpha, \tag{26}$$

where K_b denotes the number of surgeries completed in block b, assumed independent and identically distributed according to $p_s(\cdot)$.

In other words, we seek the smallest number of blocks such that the probability of completing at least Q' surgeries is no smaller than α. This procedure is applied separately to the lower and upper activity targets. Algorithm 2 summarises the computation.

Algorithm 2 Block demand estimation via successive convolutions

Input: set of subspecialties $\mathcal{S}$, targets $(\underline{Q}, \overline{Q})$, confidence level α, PMF p_s for each $s \in \mathcal{S}$
Output: block requirements $(\underline{B}_s, \overline{B}_s)$ for each $s \in \mathcal{S}$
for each subspecialty $s \in \mathcal{S}$ **do**
  if $\underline{Q} = 0$ and $\overline{Q} = 0$ **then**
    $\underline{B}_s \leftarrow 0, \overline{B}_s \leftarrow 0$
  else
    for each $Q' \in \{\underline{Q}, \overline{Q}\}$ **do**
      $n \leftarrow 1$
      $p_s^{(n)} \leftarrow p_s$
      $F_s^{(n)} \leftarrow \text{CalculateCDF}\big(p_s^{(n)}\big)$
      while $1 - F_s^{(n)}(Q' - 1) < \alpha$ **do**
        $p_s^{(n+1)} \leftarrow \text{Convolve}\big(p_s^{(n)}, p_s\big)$
        $n \leftarrow n + 1$
        $F_s^{(n)} \leftarrow \text{CalculateCDF}\big(p_s^{(n)}\big)$
      end while
      if $Q' = \underline{Q}$ **then**
        $\underline{B}_s \leftarrow n$
      else
        $\overline{B}_s \leftarrow n$
      end if
    end for
  end if
end for

This preprocessing step is essential because the OTSP is defined over block allocations rather than individual surgeries. It also provides a mechanism for propagating the uncertainty captured in the strategic layer into the tactical scheduling stage without explicitly formulating the OTSP as a stochastic optimisation model. As a result, the scheduling problem can be solved as a deterministic MILP while remaining consistent with surgery allocation targets that reflect both variability in surgery durations and the effective capacity of each block.

3.3.2. MILP Formulation

Modeling the OTSP as a MILP allows us to capture the complex interactions between subspeciality demand, block capacity, and scheduling constraints in a structured manner. To formulate the problem, we use the notation in Table 8.

The Objective Functions (OFs) are organised according to the priority levels established by the hospital's guidelines – although they can be reordered as needed to reflect different institutional priorities in other settings. To reflect these priorities, we adopt a hierarchical optimisation strategy, as proposed by Aringhieri et al. (2022). The OFs are optimised sequentially, while respecting the optimal values obtained in the previous stages.

Formally, the first OF is defined as

$$\text{Min } F_0 = \sum_{b \in \mathcal{B}} \sum_{s \in \mathcal{S}} z_{s,b} + \gamma \cdot a_b, \tag{27}$$

Table 8: Table of Symbols

Set	Description
OT	Operating theatres (both elective and outpatient)
$\mathcal{D}$	Days of the planning horizon, such that $\mathcal{D} = \{1, 2, ..., \lvert\mathcal{D}\rvert\}$
$\mathcal{M}$	Periods (morning or afternoon), such that $\mathcal{M} = \{\text{morning}, \text{afternoon}\}$
$\mathcal{B}$	Blocks, such that $\mathcal{B} = \mathcal{D} \times \mathcal{M}$
$\mathcal{S}$	Surgical subspecialties
$\mathcal{S}_{seq}$	Surgical subspecialties that necessarily require two sequential allocated blocks, such that $\mathcal{S}_{seq} \subset \mathcal{S}$
	but have high priority, such that $\mathcal{S}_{circ} \subset \mathcal{S}$
$\mathcal{S}_{prio}$	Priority surgical subspecialties, such that $\mathcal{S}_{prio} \subset \mathcal{S}$
$\mathcal{S}_{tx}$	Organ transplant surgical subspecialties, such that $\mathcal{S}_{tx} \subset \mathcal{S}$
Index	**Description**
ot	Operating theatre, with $ot \in OT$
d	Planning horizon day, with $d \in \mathcal{D}$
m	Period (morning or afternoon), with $m \in \mathcal{M}$
b	Surgery block, with $b \in \mathcal{B}$
s	Surgical subspecialty, with $s \in \mathcal{S}$
Parameter	**Description**
$\overline{B}_s$	Maximum number of blocks allocated to each subspecialty $s \in \mathcal{S}$
$\underline{B}_s$	Minimum number of blocks allocated to each subspecialty $s \in \mathcal{S}$
$NumTeam_{b,s}$	Number of teams of subspecialty $s \in \mathcal{S}$ available in block $b \in \mathcal{B}$
$NumAnest_b$	Number of anaesthetists available in block $b \in \mathcal{B}$
$NeedAnest_s$	1 if subspecialty $s \in \mathcal{S}$ requires an anaesthetist; 0 otherwise
γ	Constant to penalize the anaesthetist slack variable a_b in the objective function ($\gamma = \lvert\mathcal{B}\rvert \cdot \lvert\mathcal{S}\rvert \cdot \lvert OT\rvert$)
$\lambda_{ot,b,s}$	1 if theatre $ot \in OT$, in block $b \in \mathcal{B}$, is preferred to be by subspecialty $s \in \mathcal{S}$ in the shift block $b \in \mathcal{B}$; 0 otherwise
Variable	**Description**
$x_{ot,b,s}$	1 if theatre $ot \in OT$, in block $b \in \mathcal{B}$, is allocated to subspecialty $s \in \mathcal{S}$; 0 otherwise. This variable is only created if the allocation is feasible (i.e., specialty s is compatible with the theatre ot).
$y_{ot,d,s}$	1 when theatre $ot \in OT$, on day $d \in \mathcal{D}$, is allocated only to subspecialty $s \in \mathcal{S}$; 0 otherwise
$w_{b,s}$	Number of transplants to be performed in block $b \in \mathcal{B}$ by subspecialty $s \in \mathcal{S}$
a_b	Slack variable for the number of anaesthetists available in block $b \in \mathcal{B}$
$z_{s,b}$	Slack variable for the number of teams available in block $b \in \mathcal{B}$ for subspecialty $s \in \mathcal{S}$

which minimises the total slack in team and anaesthetist availability, with a high penalty on the latter as they are more scarce and expensive. The second function is given by

$$\text{Min } F_1 = \sum_{ot \in OT} \sum_{b \in \mathcal{B}} \sum_{s \in \mathcal{S}} x_{ot,b,s} \text{ only for variables where } \lambda_{otbs} = 0, \tag{28}$$

which minimises the number of allocations outside the aspirational theatre allocation defined for each subspecialty, thus promoting the use of preferred theatres. The third objective function is defined as

$$\text{Max } F_2 = \sum_{ot \in OT} \sum_{b \in \mathcal{B}} \sum_{s \in \mathcal{S}_{prio}} x_{ot,b,s}, \tag{29}$$

which maximises the allocation of priority subspecialties, ensuring that they receive as much capacity as possible within the constraints of the system. The fourth OF is given by

$$\text{Max } F_3 = \sum_{ot \in OT} \sum_{b \in \mathcal{B}} \sum_{s \in \mathcal{S}} x_{ot,b,s}, \tag{30}$$

which maximises the total number of allocated blocks, thus promoting overall capacity utilisation and service provision. Finally, the fifth OF is defined as

$$\text{Max } F_4 = \sum_{ot \in OT} \sum_{b \in \mathcal{B}} \sum_{s \in \mathcal{S}} x_{ot,b,s}, \tag{31}$$

which maximises the number of sequentially allocated blocks. This is particularly important for improving the logistics of the surgical process, reducing the need for equipment and team transfers between theatres.

With this, we formulate the scheduling problem with the LT-OTS(Long-Term Operating Theatre Scheduling) model, incorporating these OFs and constraints, but with the flexibility to accommodate other settings as needed.

$(LT-OTS)$ Optimize F_0–F_5

subject to

$$\sum_{ot\in OT}\sum_{b\in\mathcal{B}} x_{ot,b,s} \le \overline{B}_s \qquad \forall\, s\in\mathcal{S} \tag{32}$$

$$\sum_{ot\in OT}\sum_{b\in\mathcal{B}} x_{ot,b,s} \ge \underline{B}_s \qquad \forall\, s\in\mathcal{S} \tag{33}$$

$$\sum_{s\in\mathcal{S}} x_{ot,b,s} \le 1 \qquad \forall\, ot\in OT,\ b\in\mathcal{B} \tag{34}$$

$$\sum_{ot\in OT}\sum_{s\in\mathcal{S}} NeedAnest_s\cdot x_{ot,b,s} \le NumAnest_b + a_b \qquad \forall\, b\in\mathcal{B} \tag{35}$$

$$\sum_{ot\in OT} x_{ot,b,s} \le NumTeam_{bs} + z_{s,b} \qquad \forall\ b\in\mathcal{B},\ s\in\mathcal{S} \tag{36}$$

$$x_{ot,b,s} + x_{ot(b+1)s} - 1 \le y_{ot,d,s} \qquad \forall\, ot\in OT,\ b\in\mathcal{B},\ s\in\mathcal{S},\ d\in\mathcal{D}:\ b \text{ and } b+1 \text{ are blocks of the same day } d \tag{37}$$

$$x_{ot,b,s} + x_{ot(b+1)s} \ge 2\cdot y_{ot,d,s} \qquad \forall\, ot\in OT,\ b\in\mathcal{B},\ s\in\mathcal{S}, d\in\mathcal{D}:\ d\in\mathcal{D}:\ b \text{ and } b+1 \text{ are blocks of the same day } d \tag{38}$$

$$x_{ot,b,s} = x_{ot(b+1)s} \qquad \forall\, ot\in OT,\ b\in\mathcal{B}',\ s\in\mathcal{S}_{seq},\ d\in\mathcal{D}:\ b \text{ and } b+1 \text{ are blocks of the same day } d \tag{39}$$

$$\sum_{ot\in OT} x_{ot,b,s} = 2\cdot w_{b,s} \qquad \forall\, b\in\mathcal{B},\ s\in\mathcal{S}_{tx} \tag{40}$$

$$x_{ot,b,s}\in\{0,1\} \qquad \forall\, ot\in OT,\ b\in\mathcal{B},\ s\in\mathcal{S}$$

$$y_{ot,d,s}\in\{0,1\} \qquad \forall\, ot\in OT,\ d\in\mathcal{D},\ s\in\mathcal{S}$$

$$w_{b,s}\in\mathbb{Z}_+ \qquad \forall\, b\in\mathcal{B},\ s\in\mathcal{S}$$

$$a_b\in\mathbb{Z}_+ \qquad \forall\, b\in\mathcal{B}$$

$$z_{s,b}\in\mathbb{Z}_+. \qquad \forall\, b\in\mathcal{B},\ s\in\mathcal{S}$$

Constraints (32) and (33) impose minimum and maximum bounds on the number of allocations of each subspeciality. This is part of the integration with the strategic layer, as these bounds are integrated to the (R, Q) policy and designed to enforce a prescribed waiting time requirement subspeciality, while taking full consideration of the uncertainty in surgery durations and the associated risk of surgery cancellations. Constraint (34) ensures that each theatre and time block receives at most one subspeciality. Constraint (35) is a soft constraint that limits the number of allocations that require anaesthetists according to their availability in each block ($NumAnest_b$), while allowing for some flexibility through the slack variable a_b. Constraint (36) similarly limits the number of allocations for each subspeciality based on the availability of medical teams $NumTeam_{bs}$, with the slack variable $z_{s,b}$ providing flexibility when necessary. Constraints (37) and (38) together guarantee sequential allocation of blocks on the same day when $y_{ot,d,s} = 1$. Subspecialities requiring two consecutive blocks are specifically addressed by constraint (39). Constraint (40) ensures that transplant subspecialties are allocated in pairs, reflecting the need for simultaneous operating theatre use for donor and recipient procedures.

No explicit constraints are required to model theatre-subspecialty compatibility. Allocation variables $x_{ot,b,s}$ are introduced only for compatible pairs (s, ot), where theatre $ot \in OT$ provides the infrastructure required by subspecialty $s \in \mathcal{S}$. Hence, infeasible assignments are excluded a priori from the feasible region, and all feasible solutions automatically satisfy compatibility requirements.

4. Computational Study

We conduct a computational study to evaluate the performance of the proposed approach using real-world data from HC - Unicamp. While the theoretical results provide guarantees on system stability, it is essential to assess the practical implications in a realistic setting. To this end, we simulate the long-term surgery planning process and compare the proposed approach with a baseline policy in terms of patient waiting-lists and resource utilisation.

4.1. Real-world Data

The dataset comprises detailed information on surgery durations, operating theatre availability, team constraints, and specialty schedules over a one-year period (2025). Each month is treated as an individual instance of the OTSP to be solved in Operating Theatre Scheduling module, while the full year is used to evaluate long-term system behaviour.

4.2. Derivation of Model Inputs

The data are processed to derive the inputs required for the strategic and operational components of the proposed approach.

Demand distributions. Patient arrivals are modelled as Poisson processes. For each subspecialty, the arrival rate λ is estimated as the average monthly demand derived from historical activity from each subspecialty. Then, after building the demand distributions from λ, we use it as inputs for the Queue Management module to determine the (R, Q) policy parameters, ensuring that waiting-time requirements are met with high probability despite demand uncertainty.

Surgery duration distributions. For each subspecialty, empirical distributions of surgery durations are estimated from historical data. These distributions are used in the Overtime and Cancellation module to evaluate the newsvendor model and determine the number of surgeries scheduled per block; see Section 3.2.

Tactical parameters. All parameters of the LT-OTS model (see Table 8 in Section 3.3) are based on real-world data, i.e., the actual values observed in each month of the year 2025. These include theatre compatibilities, staff availability, and working calendars.

4.3. Experimental Setup

Experiments are conducted on a machine equipped with an Intel Core 7 150U processor (5.40 GHz) and 32 GB of RAM. All modules are implemented in Python 3.12, and the MILP model is solved using SCIP 9.2.3 with default parameter settings, interfaced through the Pyomo library.

For the Queue Management module, the waiting time target is set to $\tau = 6$ months with confidence level $\alpha = 0.1$, and the threshold R is defined as the midpoint between $\underline{Q}$ and $\overline{Q}$. In other words, we design the (R, Q) policy to ensure that, with 90% confidence (i.e., $1 - \alpha$), patients are treated within 6 months from the time they join the queue, while the threshold R is set to the average of the lower and upper bounds of the control interval, providing a balanced point for triggering scheduling decisions.

The parameters for the Overtime and Cancellation module are set as follows. The unit idle-time cost, c, is set to 1, whereas the overtime penalty cost, p, is set to $2 \cdot c$. Under this specification, and analogously to the example at the end of Section 3.2.1, the optimal time budget is given by $t^* = F_{T_D}^{-1}\left(\frac{2}{3}\right)$, which implies that at least two thirds of the green-lighted surgeries are expected to be completed without incurring overtime. In the inventory model step of this module, the marginal value of scheduling an additional surgery in a block, denoted by g, is set to 2, representing the expected gain from treating one additional patient. The cancellation cost is specified as $c(x) = x^2$, where x denotes the number of surgeries cancelled in a block, thereby capturing the increasing marginal cost of cancellations.

Baseline policy. The baseline policy excludes the strategic modules. All specialities are scheduled in every period, and the MILP model is solved using only operational constraints, without (R, Q) control or overtime optimisation. Essentially, we do so by deactivating the block demand constraints (32) and (33) in the LT-OTSmodel and using the original surgery duration distributions (without adjustment for overtime or cancellations) to drawn the number of surgeries that happened into each scheduled block. The patient arrival process and the operational parameters remain the same as in the proposed approach, ensuring a fair comparison.

4.4. Simulation Framework

Figure 4 illustrates the simulation flow. The system evolves over a monthly decision cycle. In each period, patient arrivals are realised according to the estimated demand distributions, and queues are updated accordingly. Under the proposed approach, only specialties with queue size exceeding R are selected for scheduling.

The MILP model then determines the allocation of operating theatre blocks. For the proposed approach, the number of surgeries is sampled based on the distributions obtained from the Overtime and Cancellation module. Queues are updated after the monthly cycle, and the process repeats at each new period.

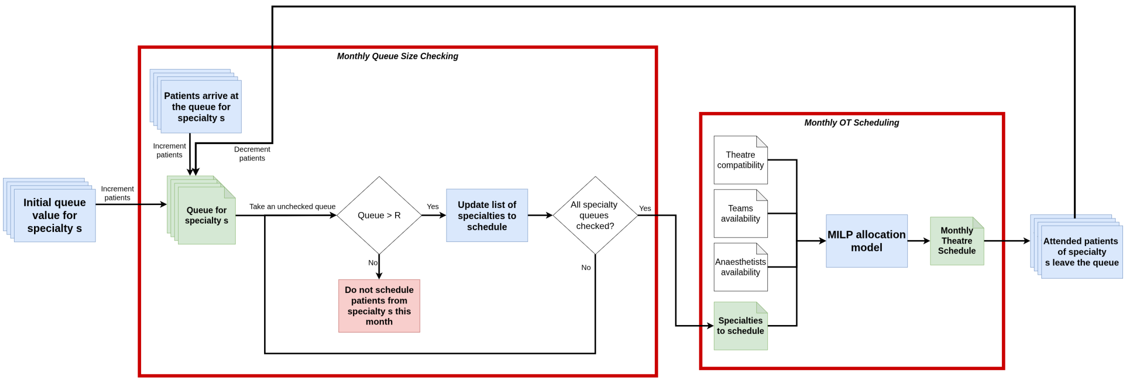


Figure 4: Long-term simulation flow.

To assess long-term stability of the framework and the baseline, we perform 100 independent simulation runs each, using different random seeds to capture the stochasticity in surgery durations and patient arrivals. This way, we can evaluate not only the expected performance but also the variability and robustness of the proposed approach compared to the baseline policy.

4.5. Results from Strategic Decision-Making

This section reports and analyses the results obtained from the strategic components of the proposed framework, namely the Queue Management and the Overtime and Cancellation modules. Although these two components are solved independently, their outputs are combined to support tactical decisions in the OT Scheduling module. This decomposition has an important computational implication: the strategic modules are computed once, and their outputs are then used as fixed inputs by the scheduling module, which is executed at each monthly cycle of the simulation horizon.

We first discuss the results of the Overtime and Cancellation module, which estimates the number of surgeries that can be planned in each block while controlling cancellation risk. We then present the results of the (R, Q) policy, which defines the control intervals for each subspecialty. These intervals are reported after conversion into block units (see Section 3.3.1), using the expected number of surgeries per block obtained from the Overtime and Cancellation module. This conversion allows the control intervals to be compared across subspecialties in a common operational unit that is directly relevant to the scheduling module. In practice, the number of patients that can be treated in a block depends on both the subspecialty-specific distribution of surgical durations and the cancellation rules defined by the Overtime and Cancellation module.

4.5.1. Overtime and Cancellation Results

This section reports the results of the Overtime and Cancellation module for Foot Orthopaedics, which is used as a representative subspecialty. The same analysis is performed for all subspecialties, with the complete results reported in Appendix **??**. Figure 5 shows the probability distribution of the number of surgeries completed in a Foot Orthopaedics block for different planned values of n. The distributions are obtained after applying the newsvendor-based cancellation rule defined in Section 3.2.1, according to which an additional surgery is performed only if the remaining session time is sufficiently large.

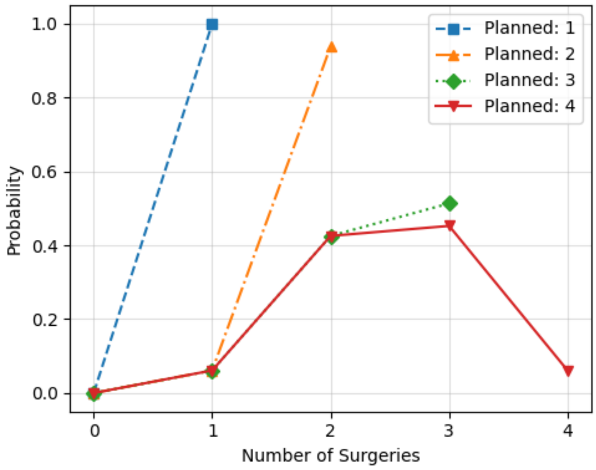


Figure 5: Probability distribution of completed surgeries in Foot Orthopaedics blocks for different planned values of n.

The results reveal a clear trade-off between block productivity and cancellation risk. When only one surgery is planned in a session ($n = 1$), the surgery is always completed and the cancellation probability is zero. Although this planning decision is operationally safe, it underuses the available block capacity for this subspecialty. When two surgeries are planned ($n = 2$), the probability mass remains highly concentrated around two completed surgeries, indicating that the block can usually accommodate this workload without substantial cancellation risk. When three surgeries are planned ($n = 3$), the distribution becomes more dispersed: completing all three surgeries remains feasible, but the probability of completing only two surgeries becomes more relevant. This indicates that the third planned surgery increases expected throughput while introducing a non-negligible cancellation risk. Finally, when four surgeries are planned ($n = 4$), the probability of completing all planned surgeries becomes very low, and the realised output is more likely to remain below the nominal plan.

These findings highlight that a theatre block should not be treated as a deterministic capacity unit capable of accommodating a fixed number of surgeries with certainty. Instead, increasing the number of planned surgeries changes the probability distribution of realised surgical output. Consequently, using only nominal capacity would overestimate the effective contribution of a block, particularly when many surgeries are assigned to the same session. This insight is central to the proposed framework: while the (R, Q) policy may indicate that a certain number of patients should be released for treatment during a monthly cycle, the Overtime and Cancellation module determines how much of this demand can realistically be assigned to each block without creating excessive cancellation risk.

Figure 6 complements this analysis by comparing the expected marginal gain and the expected cancellation cost associated with each value of n for the same subspecialty. Whereas Figure 5 reports the probability distribution of realised outputs, Figure 6 converts this uncertainty into a decision criterion for selecting the planned number of surgeries per block, as defined by the inventory-based model in Section 3.2.3.

The gain–cost comparison confirms that planning too few surgeries underuses available OT capacity, whereas planning too many surgeries produces an unfavourable cancellation profile. For $n = 1$ and $n = 2$, the expected gain is substantially higher than the expected cost, indicating that these planning levels are operationally conservative. However, they also leave potential throughput unused, particularly when compared with the additional output that can be obtained by planning a third surgery. For $n = 3$, the expected cancellation cost increases, but remains below the

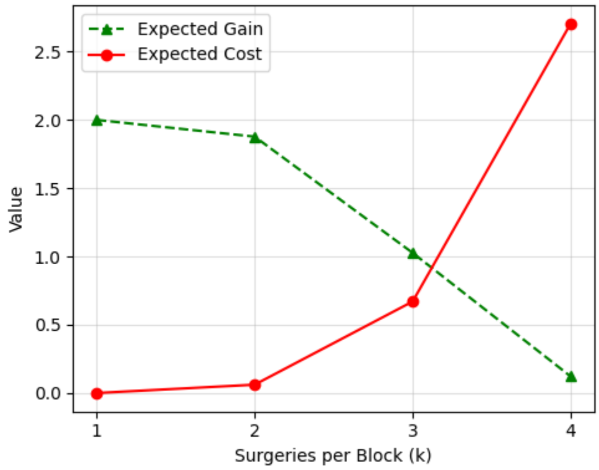


Figure 6: Expected marginal gain and expected cancellation cost associated with planning n surgeries in a Foot Orthopaedics block.

expected gain. This suggests that the third planned surgery is still justified, as it improves expected block productivity without making cancellation risk dominant.

The main change occurs when moving from $n = 3$ to $n = 4$. At this point, the expected gain drops sharply, while the expected cancellation cost increases substantially. This pattern indicates that the fourth planned surgery contributes little to the expected number of completed surgeries and mainly increases the likelihood of cancellations. Therefore, for Foot Orthopaedics, the experimental results support $n^* = 3$ as the recommended number of surgeries to plan per block. This value represents the highest planning level for which the expected benefit of adding surgeries still compensates for the associated cancellation cost.

Thus, the Overtime and Cancellation results show that the effective capacity of each block is subspecialty-specific and depends on the empirical distribution of surgical durations. For Foot Orthopaedics, three planned surgeries per block provide the best balance between productivity and cancellation risk. This output is then combined with the (R, Q) limits to determine how many blocks should be allocated to this subspecialty in each monthly cycle. Therefore, the MILP does not allocate blocks solely on the basis of nominal session availability. Instead, it uses a probabilistic estimate of how many surgeries each allocated block can realistically deliver while controlling overtime-related cancellations.

4.5.2. (R, Q) Batch Size Results

This section reports the control intervals obtained from the (R, Q) policy for all subspecialties. Figure 7 summarises these intervals after conversion into block units (see Section 3.3.1). To improve readability, subspecialties are ordered according to interval width, $\overline{Q} - \underline{Q}$, rather than alphabetically.

The results reveal substantial heterogeneity across subspecialties. Although most services are associated with relatively narrow feasible intervals, a smaller group exhibits considerably higher lower and upper limits. This pattern is consistent with variation in patient demand distributions. Subspecialties with higher expected demand tend to produce higher values of both $\underline{Q}$ and $\overline{Q}$, whereas higher demand variability tends to generate wider intervals.

High-interval subspecialties, such as ENT (Ear, Nose and Throat), NEURO (Neurosurgery), and VASC (Vascular Surgery), provide greater flexibility because the number of blocks, and consequently the number of patients released for scheduling, can vary substantially from one planning cycle to another without compromising waiting-time requirements. In a high-demand public hospital, this flexibility is particularly important for broad-access or high-referral subspecialties, which may accumulate patients rapidly when surgical capacity is constrained. At the same time, these subspecialties may release significant capacity to other services when their own queues are under control.

By contrast, low-interval subspecialties operate within tighter allocation ranges. This does not imply lower clinical importance. Rather, it indicates that their demand is smaller and more stable from the perspective of queue replenishment. Subspecialties in the medium group occupy an intermediate position: their demand is recurrent enough to

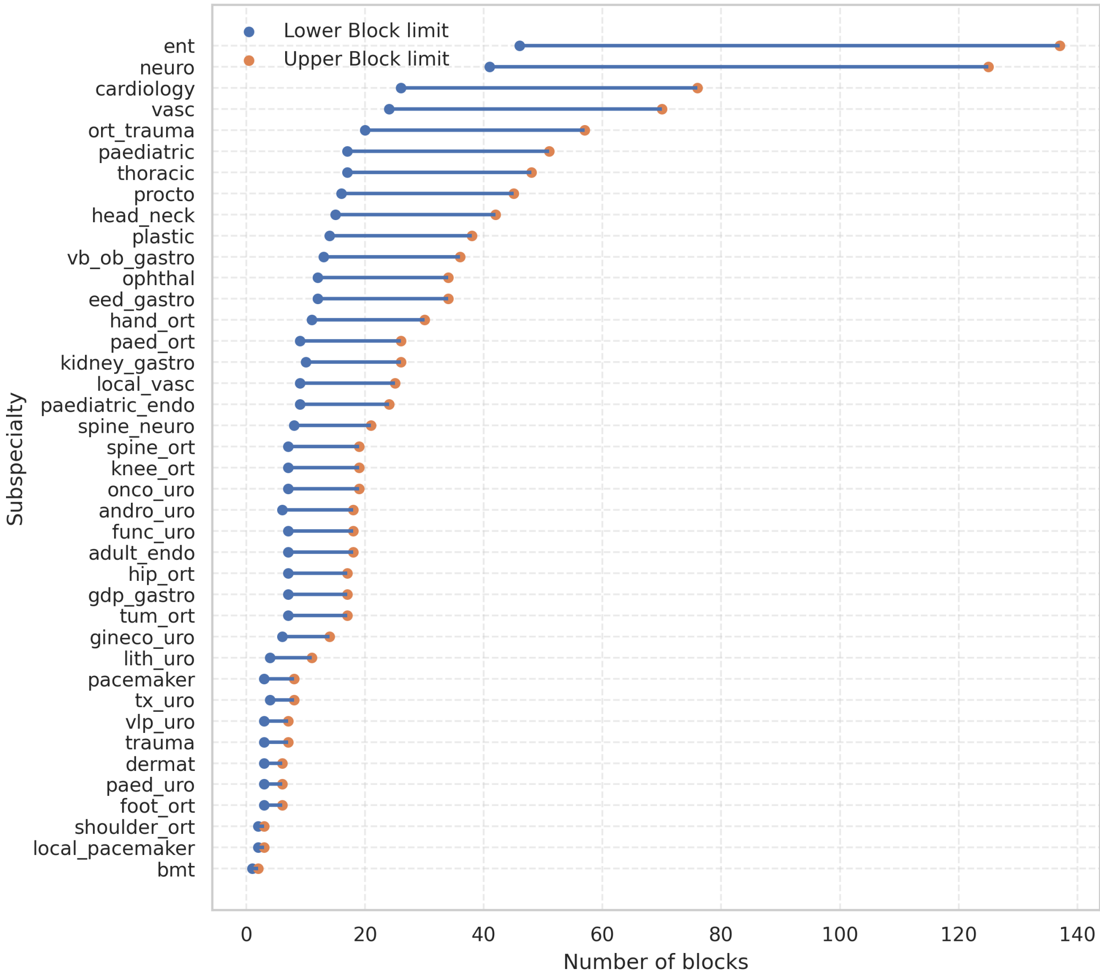


Figure 7: Control intervals for the (R, Q) policy across subspecialties. The intervals represent the admissible range of blocks that can be allocated to each subspecialty when its queue exceeds the reorder point R.

require some flexibility, but not sufficiently large or variable to dominate the release process.

These results have direct implications for capacity management planning. Because the (R, Q) limits determine the number of patients transferred from the queue-management layer to the tactical planning stage, subspecialties with wider and higher intervals are more likely to influence capacity allocation and overtime requirements. This does not necessarily mean that these services should always receive more operating-theatre time. Instead, it indicates that their larger and more variable demand must be explicitly represented in the tactical optimisation problem. Conversely, subspecialties with narrow intervals generate smaller and more predictable demand releases, reducing their impact on capacity allocation decisions.

It is also important to recall that the (R, Q) policy triggers allocation decisions for a subspecialty only when its queue size exceeds the threshold R. Thus, even subspecialties with wider intervals will not necessarily dominate the block allocation process, because they are considered for scheduling only when their queues are sufficiently large. This feedback mechanism ensures that the system remains responsive to actual demand conditions, rather than being driven solely by potential variability in patient arrivals.

Overall, the control intervals provide a quantitative link between queue dynamics and tactical planning. They translate subspecialty-specific demand characteristics into structured release limits, ensuring that the scheduling module receives adaptive demand inputs that are consistent with both the statistical behaviour of the queues and the operational characteristics of a high-demand public hospital such as HC - Unicamp.

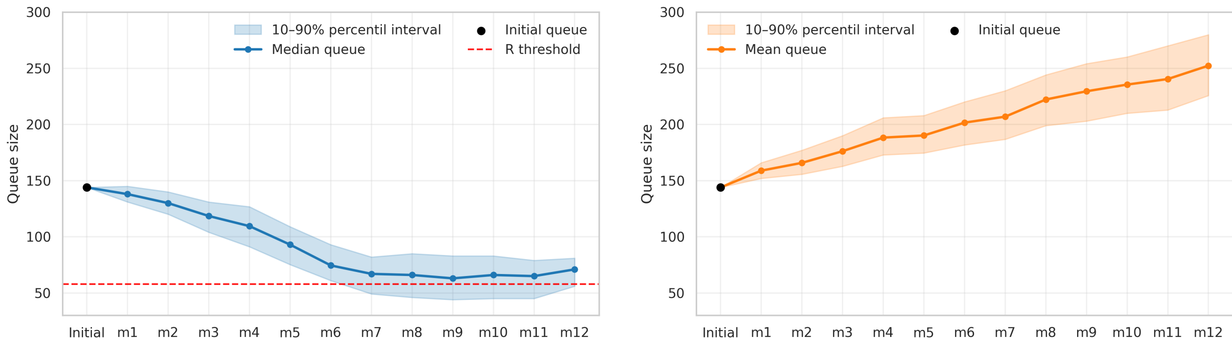
4.6. Long-term Stability Analysis

This section evaluates the long-term stability of the system under the proposed framework and compares it with the baseline policy. The analysis focuses on two complementary dimensions: the evolution of patient queues over time and the monthly allocation of operating-theatre capacity. Together, these results allow us to assess whether the proposed framework not only reduces individual queues, but also supports stable and demand-responsive planning at the system level.

4.6.1. Queue Management Results

The one-year simulation horizon provides a detailed view of queue dynamics under both policies. Results are reported using the 10–90% percentile interval to characterise variability across simulation replications.

Figure 8 presents the queue evolution for *Upper Digestive Endoscopy (UDP) – Paediatric*. Under the proposed framework (Figure 8a), the system exhibits two distinct phases. The first is a transient phase, during which the queue decreases steadily from its initial level. This is followed by a controlled regime, in which the queue starts to stabilise, approximately between months 6 and 9, and then fluctuates around the reorder threshold R. This transition reflects the action of the (R, Q) policy, which introduces a demand-driven feedback mechanism: allocation decisions are triggered only when the queue exceeds R, thereby preventing unnecessary service once the waiting list has been reduced. As a result, the system converges towards a stable operating region that balances waiting-time requirements with capacity constraints.



(a) Queue evolution under the proposed framework.

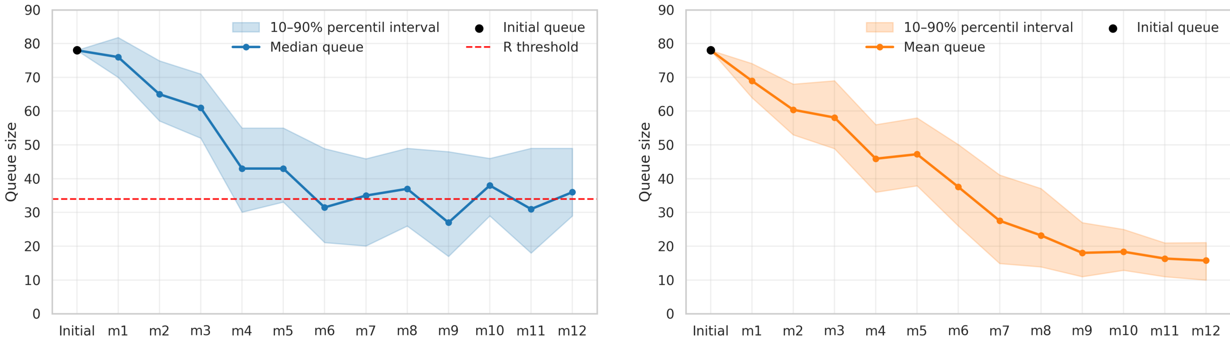
(b) Queue evolution under the baseline policy.

Figure 8: Evolution of patient queues over time for the subspecialty *UDP – Paediatric*.

In contrast, the baseline policy leads to a markedly different trajectory (Figure 8b). The queue grows throughout the simulation horizon and is accompanied by increasing dispersion across replications. This pattern suggests a persistent mismatch between demand and allocated capacity for this subspecialty. Because the baseline allocation rule does not explicitly account for the current queue state, capacity may be assigned when it is not most needed, while queues requiring additional intervention continue to accumulate.

A similar stability pattern is observed for *Gastroenterology – Esophagus-Stomach-Duodenum (ESD)* in Figure 9. Under the proposed framework (Figure 9a), the initial waiting list is reduced rapidly and the queue subsequently remains close to the reorder threshold. This indicates that the (R, Q) mechanism adapts allocation decisions to the evolving state of each subspecialty while maintaining a consistent target operating region. Importantly, the queue does not continue decreasing indefinitely. Instead, it stabilises around R, which is the intended behaviour of a reorder-point policy designed to regulate, rather than eliminate, queues.

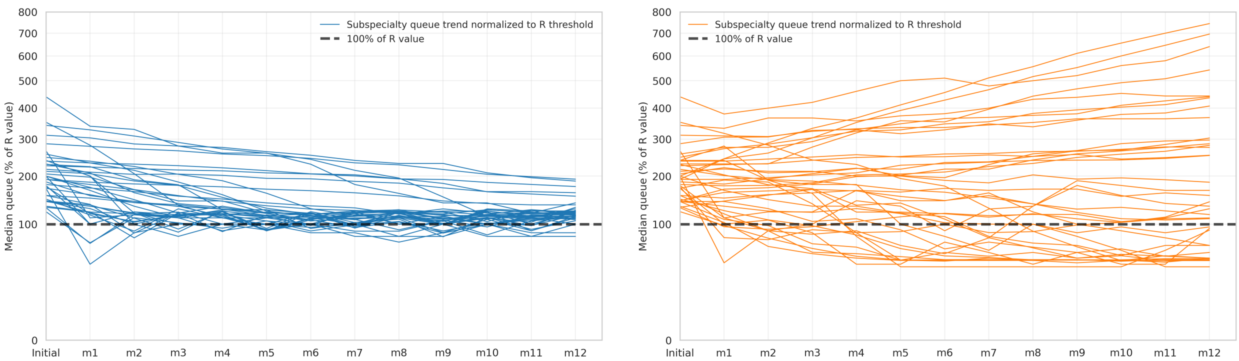
For the baseline policy (Figure 9b), the queue also decreases over time. However, this local improvement should not be interpreted as evidence of system-wide stability. As the baseline does not prioritise subspecialties according to queue status, it may allocate capacity to services whose queues are already manageable, while other subspecialties remain congested. Thus, the baseline can produce favourable trajectories for selected queues while still generating



(a) Proposed framework. (b) Baseline policy.

Figure 9: Evolution of patient queues over time for the subspecialty *Gastroenterology – Esophagus-Stomach-Duodenum (ESD)*.

inefficient allocation patterns at the system level. This distinction is important: analysing a single subspecialty in isolation may obscure whether a policy is effectively coordinating capacity across the full portfolio of surgical queues.



(a) Proposed framework. (b) Baseline policy.

Figure 10: Median queue trajectories for all subspecialties over the simulation horizon, normalised by the corresponding reorder threshold R. The dashed line represents 100% of the R value.

Figure 10 complements the subspecialty-level analysis by reporting the median queue trajectories for all subspecialties, normalised by their corresponding reorder threshold R, i.e., $\frac{Q(m)}{R} \cdot 100$, where $Q(m)$ is the queue length in monthly cycle m. Although the baseline policy do not use the (R, Q) policy, normalising queues by R provides a common reference point for comparing the two approaches across subspecialties. Also, this normalisation enables comparisons across subspecialties with different demand profiles and queue scales.

Under the proposed framework (Figure 10a), queues progressively concentrate around the 100% reference line, indicating convergence towards the intended controlled region. This system-wide pattern shows that the proposed framework regulates queues consistently across subspecialties. As a consequence, waiting-time requirements are met with high probability, because queues are maintained within the limits defined by the (R, Q) policy. Specifically, queues neither grow indefinitely nor remain persistently above their reorder thresholds. This indicates that an arriving patient has a high probability of being treated within the target waiting time, as the policy prevents excessive patient accumulation.

In contrast, the baseline policy (Figure 10b) exhibits a more heterogeneous and unstable pattern. Some subspecialties move towards lower queue levels, whereas several others experience sustained growth and remain substantially above their reorder thresholds. This dispersion reveals the absence of a system-wide stabilising mechanism. The baseline may perform adequately for selected subspecialties, but it does not ensure convergence across the full portfolio of surgical queues. More importantly, it may fail to meet waiting-time requirements for many subspecialties, because queues are not explicitly controlled.

By driving queues towards a controlled operating region, limiting excessive waiting-list accumulation, and supporting demand-driven prioritisation, the proposed framework provides meaningful long-term stability, which is a critical requirement for high-demand public hospitals. In contrast, the baseline policy lacks an explicit feedback mechanism based on queue status. This can lead to non-compliance with waiting-time requirements and to outcomes that are locally acceptable but globally inefficient. Detailed queue trajectories for all subspecialties are provided in Appendix **??**.

4.6.2. Long-term OT Allocation Results

This section analyses the long-term allocation decisions produced by the OT Scheduling module described in Section 3.3. Figure 11 reports monthly operating-room occupancy under the proposed framework and the baseline policy. Both approaches generate feasible allocations with substantial use of available capacity. This is consistent with the MILP formulation, which maximises block allocations while respecting theatre availability, subspecialty compatibility, and monthly capacity limits. Specifically, for the proposed framework, this result indicates that the scheduling layer is able to convert planned surgical demand into executable theatre allocations. In other words, the MILP successfully implements the strategic decisions produced by the queue-control and overtime modules, translating them into operationally meaningful theatre plans.

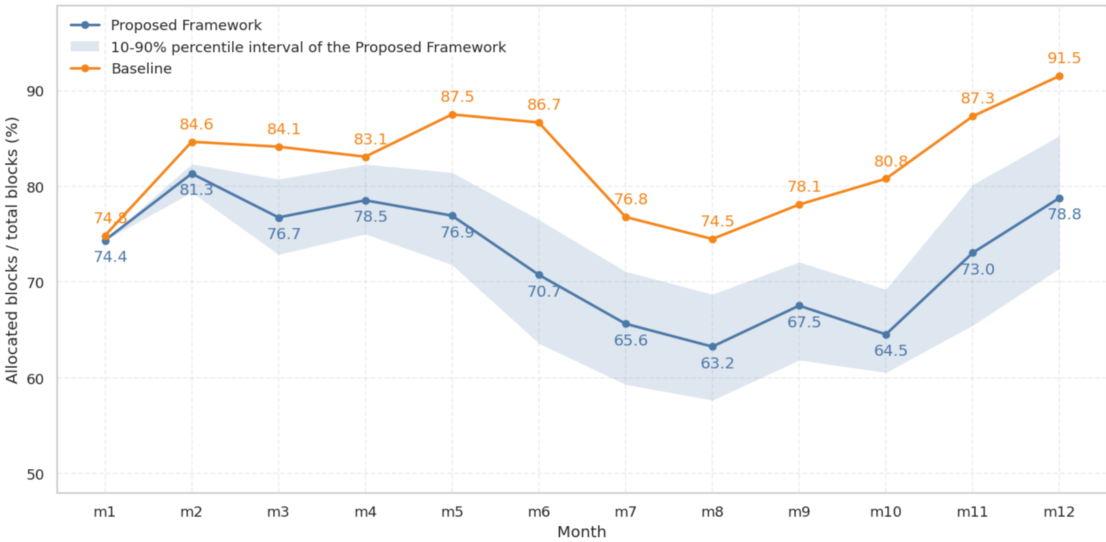


Figure 11: Monthly operating-room occupancy under the proposed framework and the baseline policy. Occupancy is calculated as the number of allocated blocks in each month divided by the total number of available blocks. The baseline policy has no shaded area because its allocation decisions are independent of realised demand and queue states. As it attempts to allocate blocks to all subspecialties in every month, the resulting occupancy is deterministic across replications.

The baseline policy systematically presents higher occupancy than the proposed framework. This difference is expected and reflects the distinct logic behind the two allocation policies. The baseline allocates capacity without explicitly considering whether each subspecialty queue is above or below its reorder point. As a consequence, it tends to distribute theatre blocks across a larger set of subspecialties and maintains high occupancy even when some queues do not require immediate intervention.

By contrast, the proposed framework triggers allocation only when a subspecialty queue exceeds its reorder threshold R. When the queue is below this level, the framework does not force an allocation merely to increase theatre utilisation. Therefore, the lower occupancy observed in some months should not be interpreted as an inefficiency of the scheduling model. Rather, it reflects a deliberate control mechanism that avoids allocating capacity to subspecialties already operating within the desired region.

Figure 12 supports this interpretation by showing the number of subspecialties allocated each month. The baseline assigns blocks to a relatively stable and high number of subspecialties, ranging from approximately 34 to 39 throughout the simulation horizon. In contrast, the proposed framework allocates fewer subspecialties in most months, with a more pronounced reduction in the second half of the year — which is consistent with the initial waiting-list sizes of each subspecialty: we draw 6 months of demand as the initial queue, so it is expected that a more significant reduction in the number of allocated subspecialties occurs after month 6, as more queues enter the controlled region. This analysis confirms that the difference in occupancy is directly associated with the queue-control mechanism: as more queues enter the controlled region, fewer subspecialties require active allocation.

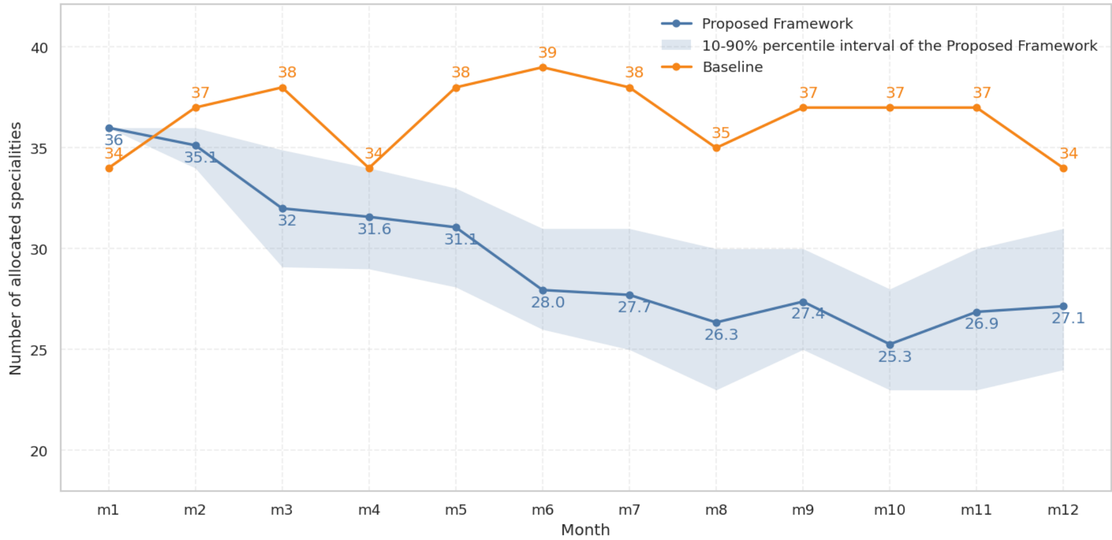


Figure 12: Number of subspecialties allocated per month under the proposed framework and the baseline policy. The baseline policy has no shaded area due to the same reason in Figure 11: its allocation decisions are independent of realised demand and queue states. As it attempts to allocate blocks to all subspecialties in every month, the resulting occupancy is deterministic across replications.

The behaviour observed in Figure 12 has important implications for long-term planning. A policy that allocates blocks to all, or nearly all, subspecialties may produce higher utilisation rates, but it can also consume capacity on queues that are already under control while neglecting those that require intervention. The proposed framework introduces a more selective allocation logic: capacity is concentrated on subspecialties whose queues require intervention, while subspecialties below R can temporarily remain unallocated. This creates a controlled form of flexibility, allowing the OT scheduling MILP to redirect blocks over time according to the evolving state of the system rather than reproducing a fixed allocation pattern.

This distinction is central to the interpretation of the occupancy results. The baseline tends to preserve regular allocation across subspecialties, leading to higher occupancy but weaker responsiveness to queue conditions. The proposed framework links allocation decisions to queue status: it may increase, reduce, or temporarily suppress allocations depending on whether each subspecialty requires intervention. As a result, the objective is not simply to fill the OT schedule, but to allocate theatre capacity where it contributes most to maintaining queue stability. In other words, the proposed framework prioritises (and indeed promotes) long-term stability and responsiveness over short-term occupancy maximisation.

5. Conclusions and Future Directions

This paper proposes a modular framework that integrates strategic and tactical decisions for elective surgery planning under uncertainty. The framework combines waiting-list control, cancellation risk management, and tactical operating theatre allocation within a coherent decision process. Rather than solving a single large-scale holistic model, the proposed approach decomposes the problem into interconnected modules. This design is particularly relevant in practice, as it allows decision-makers to adapt parameters to diverse hospital contexts or utilise a subset of the modules, while simultaneously reducing the computational burden by solving smaller, more specialised optimisation problems.

Module 1 (Queue Management) establishes waiting-list control policies for each surgical group, defining target volumes of patients to be scheduled in each planning cycle. As part of the framework's strategic layer, this module ensures long-term queue stability and supports compliance with waiting-time requirements. Module 2 (Overtime and Cancellation), also strategic, employs a newsvendor-based approach to define a time budget for each surgical block, explicitly addressing the trade-off between theatre capacity utilisation and the risk of overtime or cancellations due to stochastic surgery durations. The distributions derived from this model inform an inventory-based module to determine the optimal number of surgeries to be scheduled per subspecialty. Finally, Module 3 (Operating Theatre Scheduling) performs the tactical allocation of theatre blocks to subspecialties, using the decisions and parameters generated by the strategic modules as inputs.

Although the third module is formulated as a MILP model and solves the scheduling problem efficiently, the framework's validity does not depend on this specific modelling choice. Other exact, heuristic, or metaheuristic methods could be substituted, as the primary contribution lies in the integration of the modules and the consistency between strategic and tactical layers. This highlights the flexibility of the approach; the Operating Theatre Scheduling module can be tailored to different optimisation methods depending on the application context, problem scale, or managerial preferences, while still leveraging the strategic insights provided by the initial modules.

Beyond the theoretical contribution, the framework is evaluated through a real-world case study in a large Brazilian public referral hospital. Results from the Overtime and Cancellation module indeed provide a quantitative assessment of the trade-off between scheduling additional surgeries and exposing the system to higher cancellation risks. The findings indicate that treating surgery durations as stochastic rather than deterministic is crucial for realistic planning. Notoriously, when the number of planned surgeries per block increases, the probability of completing all scheduled surgeries decreases, and the risk of cancellations rises significantly. By using this insight to determine the effective number of surgeries to be planned in a block, we achieve strategic demand targets consistently along the planning horizon. Therefore, treating block capacity as deterministic is unrealistic, as it substantially inflates the probability of overtime.

The modified (R, Q) control policy also played a pivotal role in the framework's performance. The target batch sizes defined for each subspecialty in the case study provides the necessary flexibility to absorb demand fluctuations. It avoids both excessive queue growth, which leads to patient distress, and excessively short queues, which result in the underutilisation of costly surgical resources.

The long-term simulation results further complement these results from the strategic modules, highlighting the necessity of integrating them to tactical decisions. When tactical theatre scheduling is considered in isolation, ignoring waiting-list dynamics and duration uncertainty, the resulting plans may appear attractive regarding short-term utilisation but perform poorly in practice. For instance, under the baseline policy, many waiting lists grew uncontrollably in the experiment. In contrast, the proposed framework drives waiting lists towards their respective reorder points R and maintains stability over the simulation horizon. Furthermore, the results demonstrate that high theatre occupancy should not be interpreted in isolation: a policy may show high utilisation but allows queues to deteriorate. Conversely, in the proposed framework, lower average occupancy in certain periods reflects the fact that, once a target queue level is reached, capacity of the respective subspecialty can be redirected or held in reserve without compromising waiting-time targets. From a managerial perspective, this allows hospital administrators to identify theatre blocks that may be reallocated to other surgical groups, emergency demand, or maintenance, while preserving control over cancellations.

This study has limitations that suggest avenues for future research. First, demand was modelled as a Poisson process based on historical data. It does not invalidate the framework, as the strategic modules can be adapted to any alternative demand models, but future applications that have available data about it could incorporate even more realistic patient arrivals. Second, the framework could be extended to include bed-management decisions, both upstream and downstream. Integrating elective surgeries decisions with ward and ICU availability would help avoid situations in which planned surgeries are assigned to a subspecialty, but patients cannot be operated on because the required beds are unavailable. Third, a comprehensive sensitivity analysis could identify the impact of varying waiting-time requirements or different cost structures on the trade-off between throughput and cancellation risk. Finally, the framework could incorporate an operational decision layer to address the sequencing of surgeries within blocks, the allocation of specific surgical teams, and the real-time management of emergency cases on the day of surgery.

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